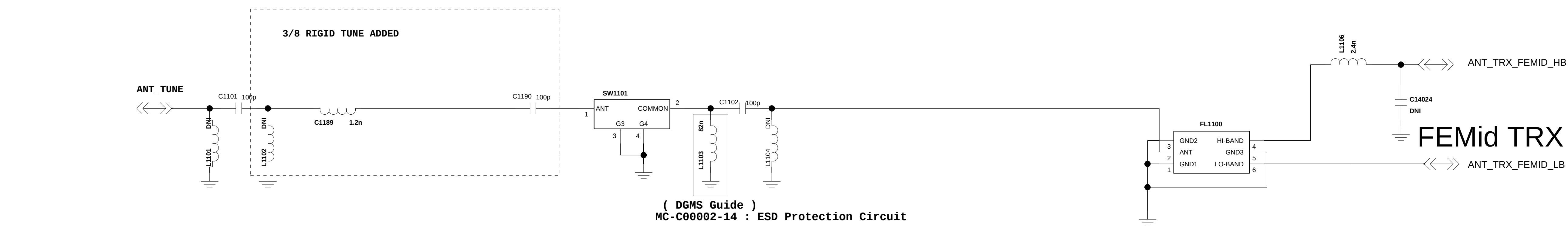


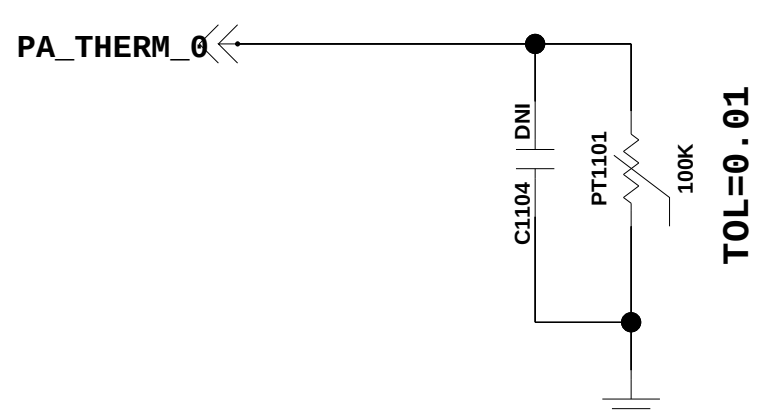
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ANT1 (LTE 1, 3, 5, 7, 8, 17 / WCMDA 1, 2, 5 / MIMO 7/ GSM)

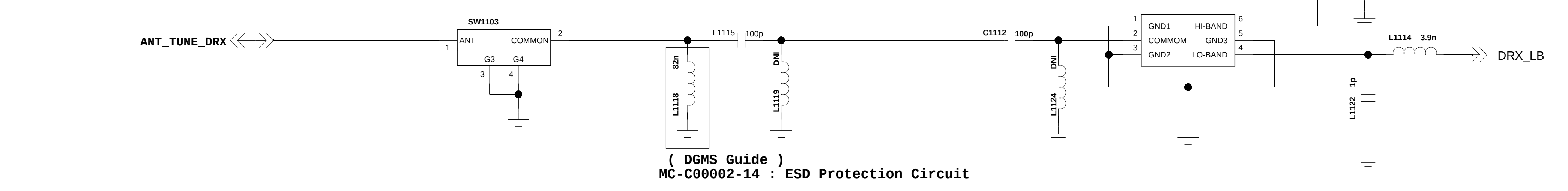
Rev_0.1



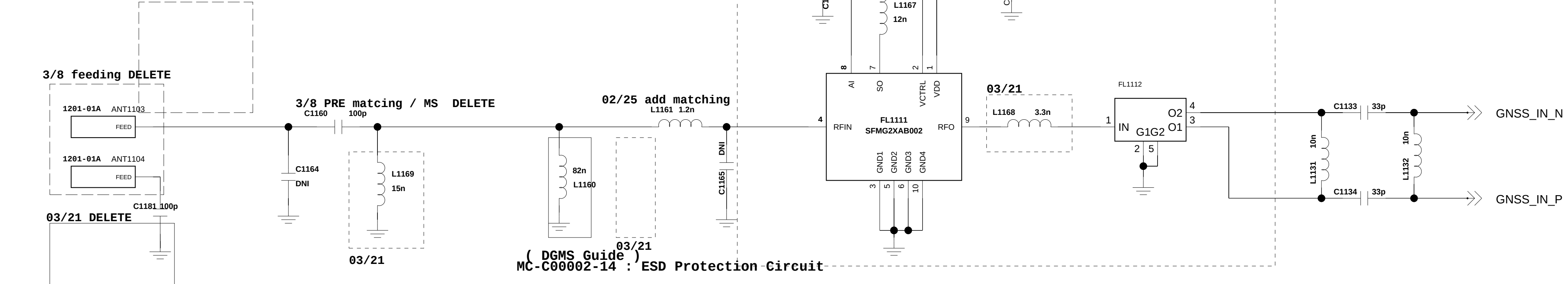
PA THERM_0



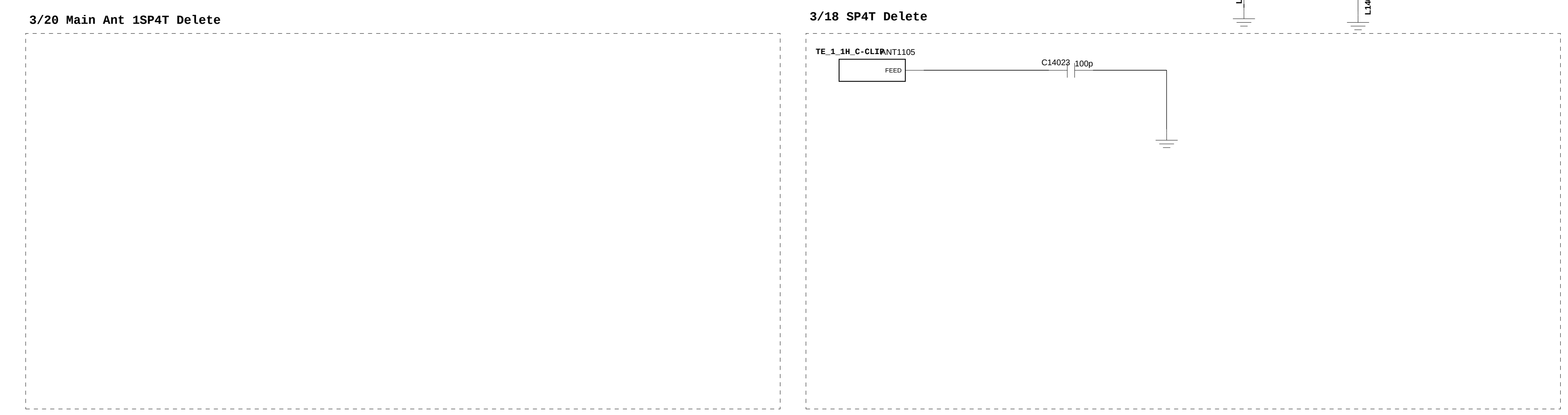
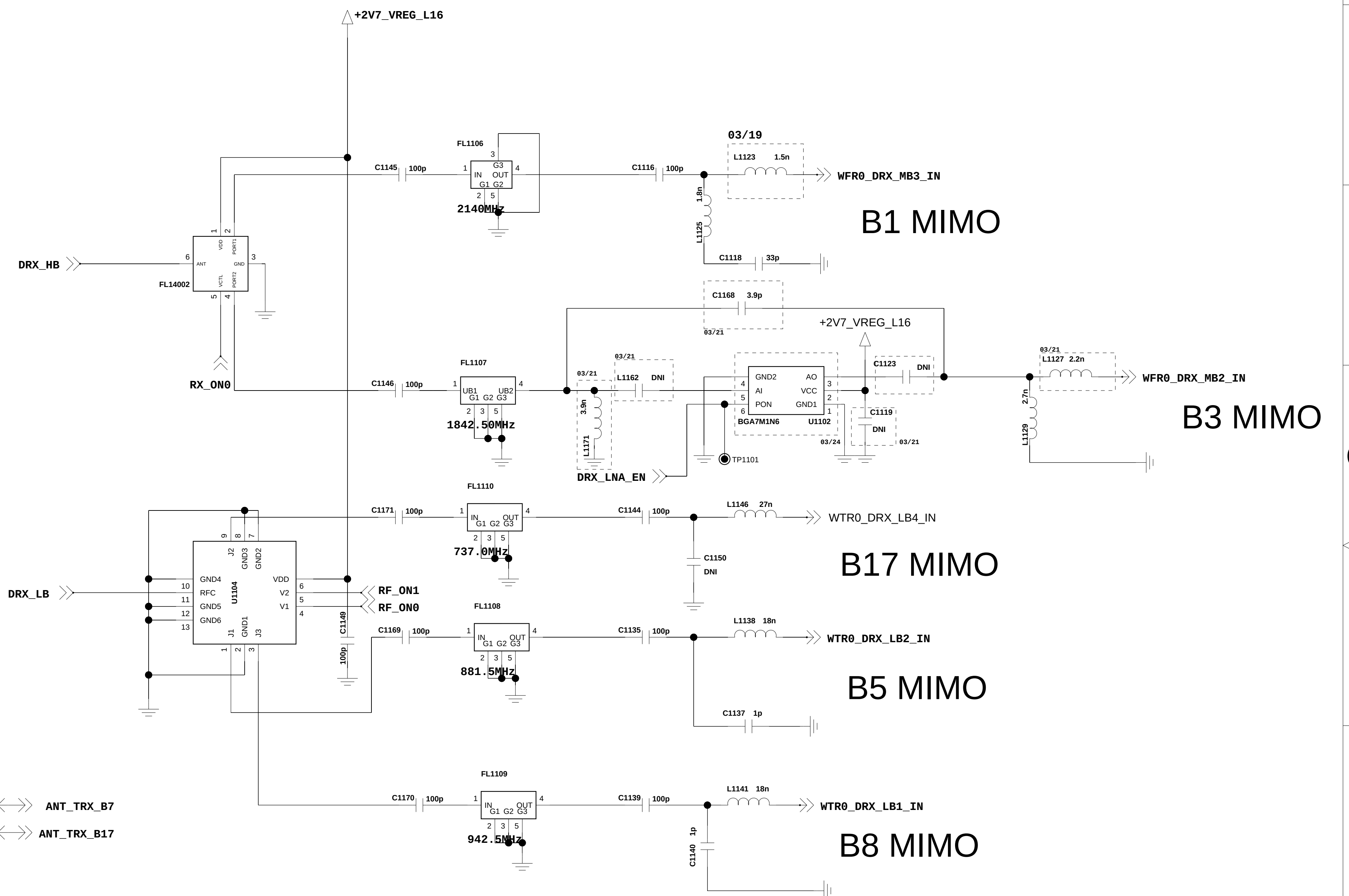
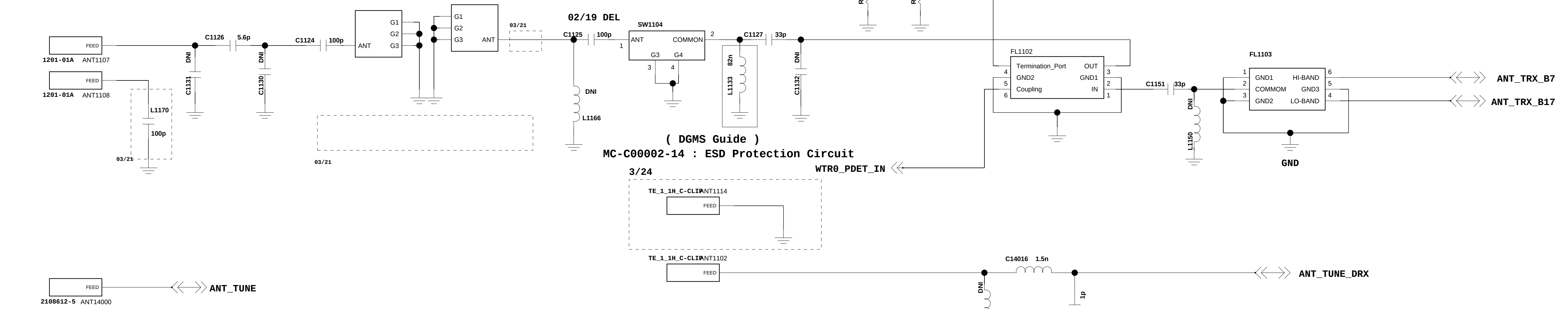
ANT3 (MIMO B1, 3, 5, 8, 17)



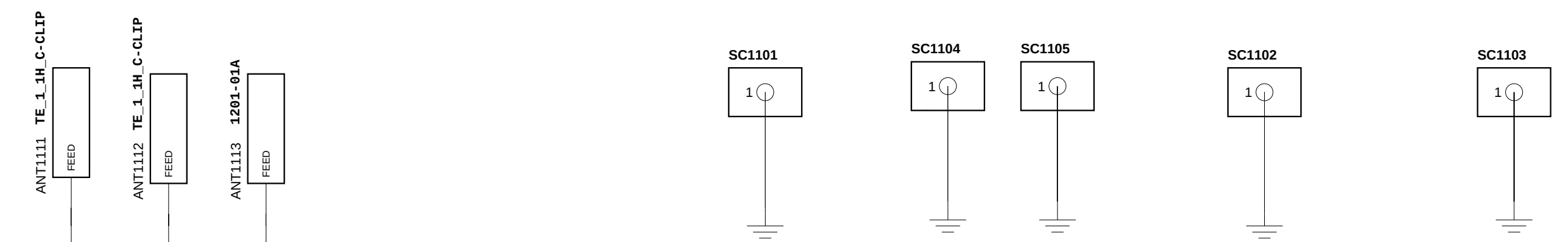
ANT5 (GNSS)



ANT2 (B7, 17)



< SUS plate >



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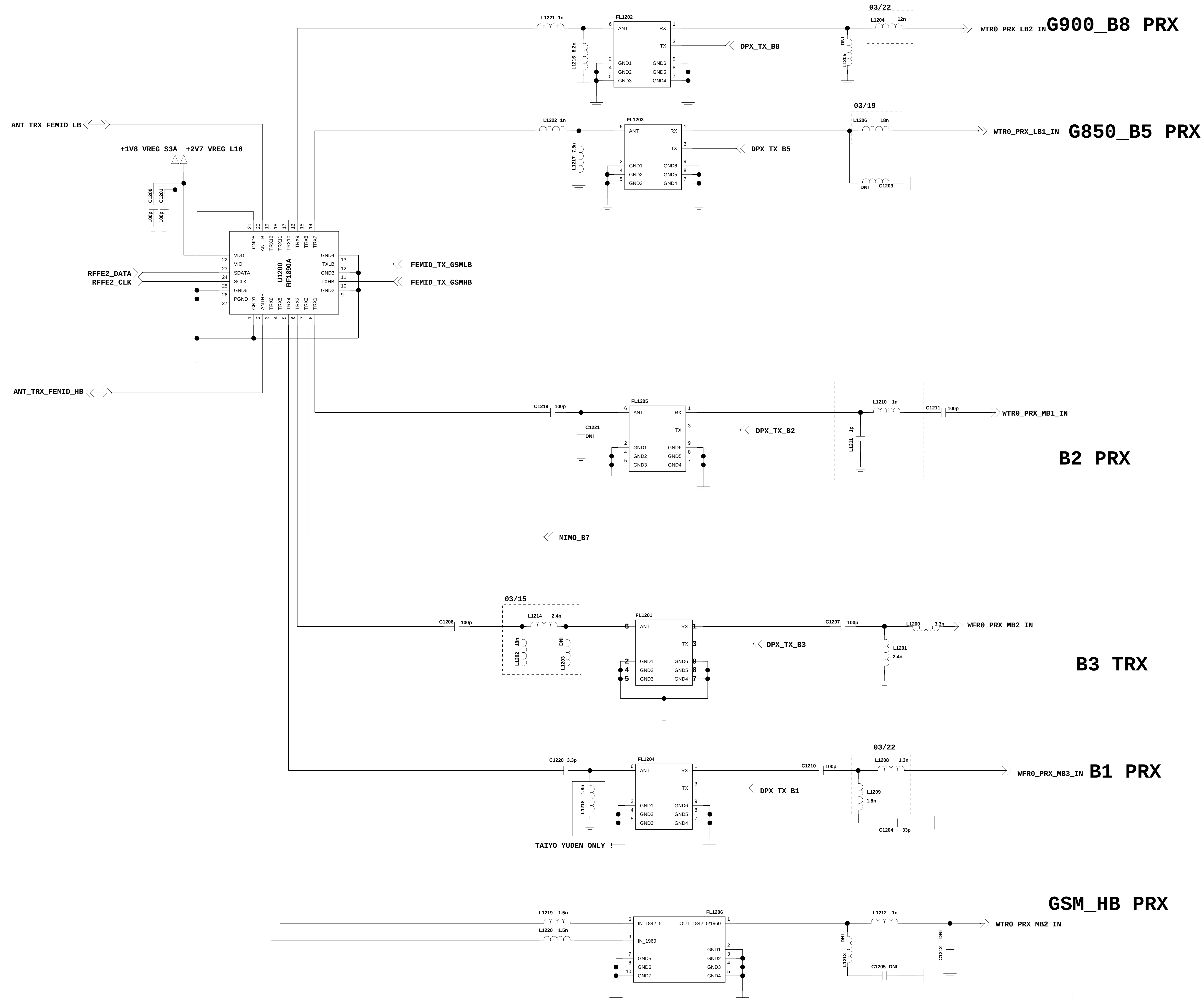
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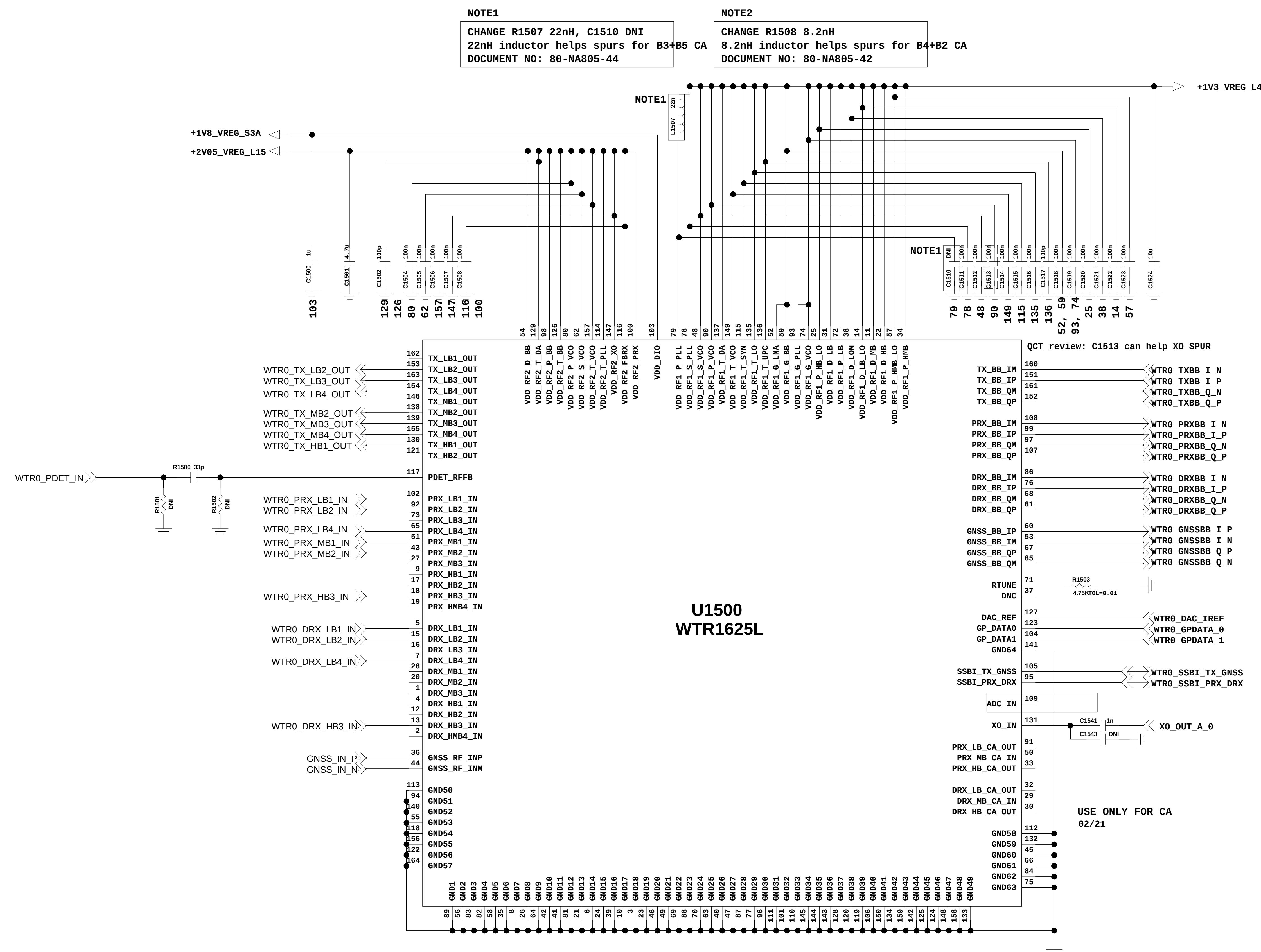
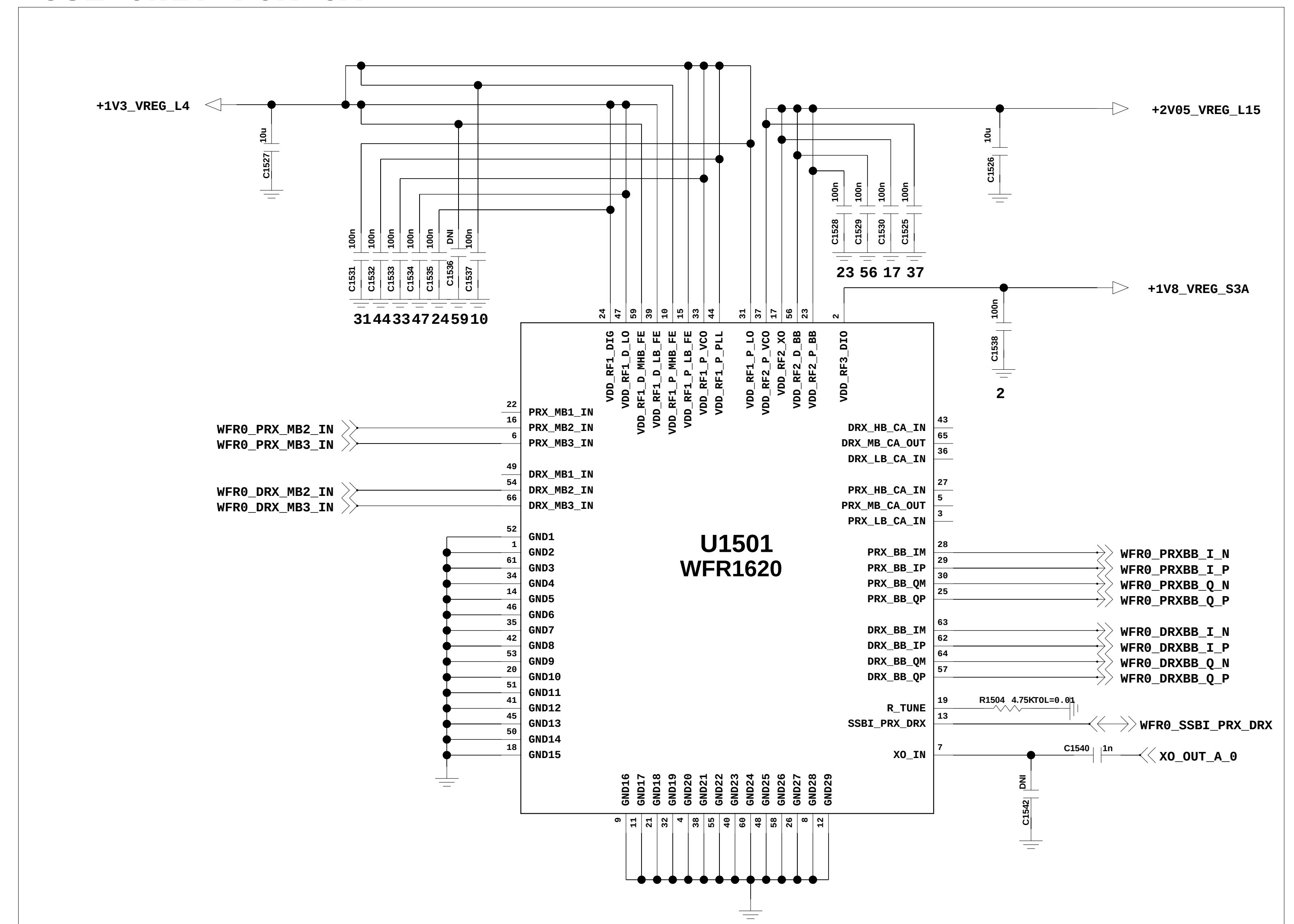
Rev_0.3



< 1-3-5-1_RFIC_CA_nCA_WTR1625 >

Rev_0.4

CAUTION: YOU MUST CHECK & MODIFY RF PORT AND GPIO ON YOUR SCHEMATICS.

**USE ONLY FOR CA**

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I2C Pull-Up



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<2-1-2-2-3_MSM8974AC_POWER>

CCDS CARD Information	
Release Date	2013.08.26
Based on Reference Schematic	Rev.M
89-NA437-41_MSM8974-MSM8974AB_BASEBAND_REFERENCE_SCHEMATIC_REV.M	

NOTE 2:QMC Checklist Rev.C
1 μ F 1 μ F cap each close to the following pin or pin group: (BD38, BC37), BF28

NOTE 1:QMC Reference Schematic Rev.L
PLACE S2B, S4B, S5B, S6B, S7B, S8B
OUTPUT CAPACITORS CLOSE TO MSM

NOTE 7:QMC Checklist Rev.C
3 μ F 1 μ F cap, one cap each close to pin F2, P4, K4

NOTE 9:QMC Checklist Rev.K
VSENSE_KRAIT_0P9, REMOTE_GND_SNS
VSENSE_KRAIT_0P9 and REMOTE_GND_SNS
need to be routed differentially

NOTE 8:QMC Reference Schematic Rev.L
PLACE S2B, S4B, S5B, S6B, S7B, S8B
OUTPUT CAPACITORS CLOSE TO MSM

NOTE 3:QMC Chekclist Rev.C
If HSIC is used, connect VDD_P4 to 1.2 V (VREG_L1_1P2)
If GPIO144/145 is used, connect VDD_P4 to 1.80 V(VREG_S3A_1P8)

NOTE 4:QMC Checklist Rev.C
VREG_S1B_0P95
Place a 0 Ω resistors between VREG_S1B_0P95 and the VDD_EBI CDC and VDD_EBI PLL pins are to star connects to VDD_MX rather than just short to a plane.
The resistor forces a separate netname and a separate routing and does not allow shorting to the plane.
We recommend to keep the resistors in the design during the routing stage to achieve the desired isolation. Once routing is complete customer can remove the resistor and simply short the isolated net back to the main plane at the location of the resistor.

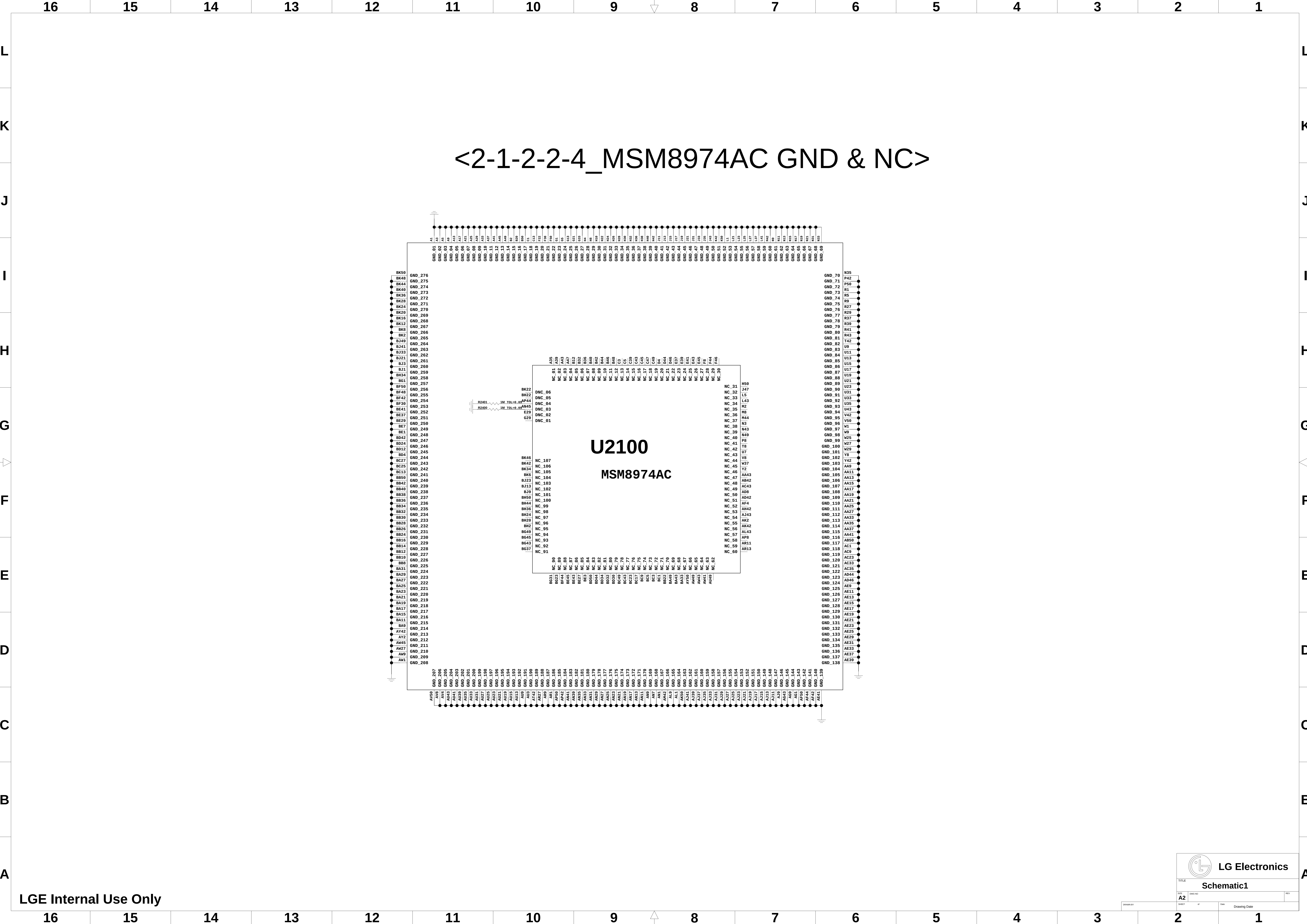
NOTE 10:QMC Reference Schematic Rev.K
PLACE S2B, S4B, S5B, S6B, S7B, S8B
OUTPUT CAPACITORS CLOSE TO MSM

U2100 MSM8974AC

LG Electronics	
Schematic1	
A2	PDM NUMBER
2-1-2-2-3_MSM8974AB_POWER	Rev

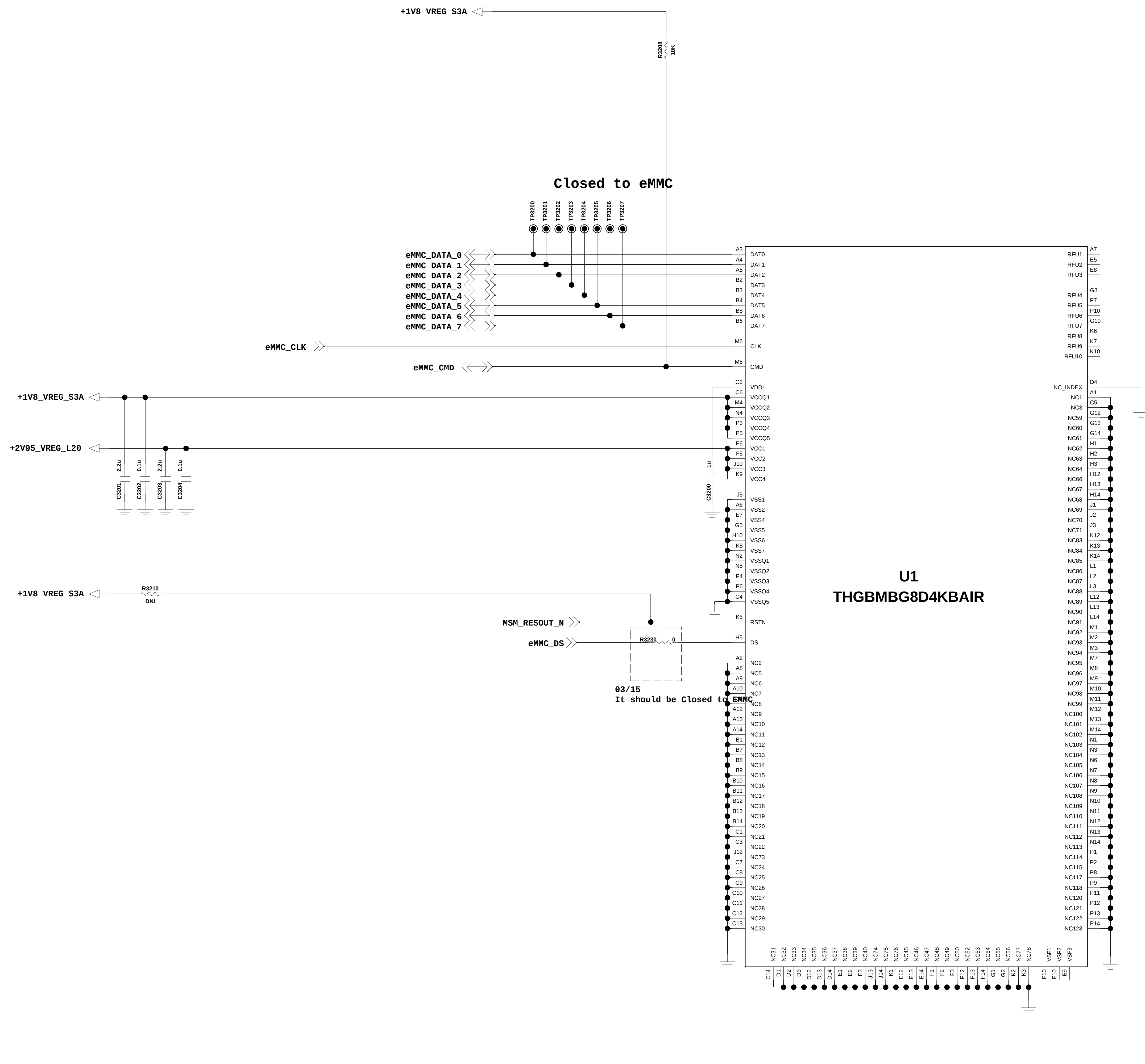
NOTE 5:QMC Checklist Rev.K
1 μ F 1 μ F cap each close to the following pin or pin group (pins in a bracket is a group and share one 1 μ F cap): (AE35, AG15, N23), (AP48, AE43, AF46), U47, (AR29, AU29)

NOTE 6:QMC Checklist Rev.K
VREG_S2B_0P9
Place a 0 Ω series resistor between VREG_S2B_0P9 and pins VDD_SDC CDC, VDD_PLL1 star route to the VDD_CORE plane. Do not short the VDD_SDC CDC, VDD_PLL1 pins directly to the main VDD_CORE plane.

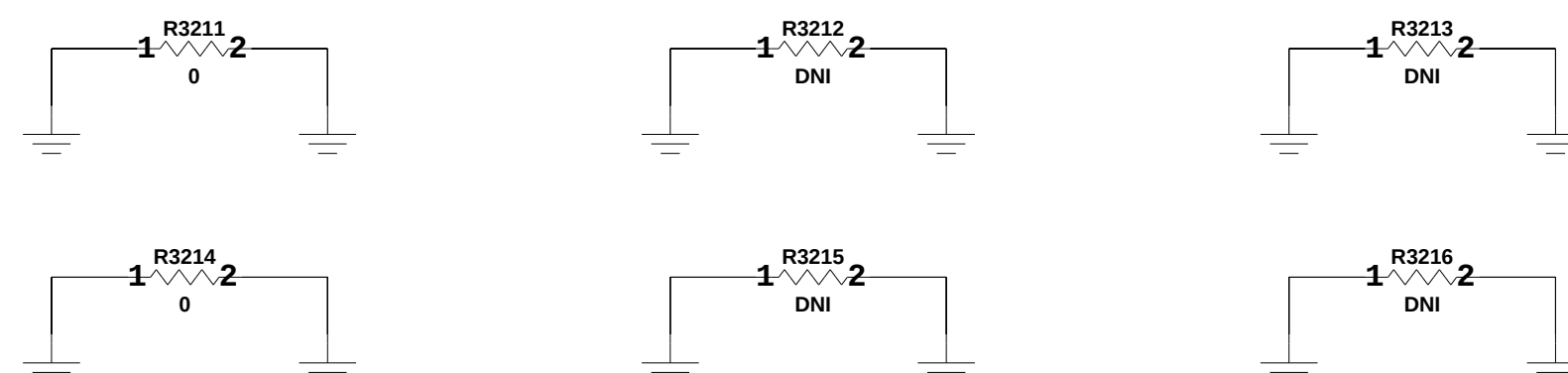


<3-2-2-3-1_eMMC_5_0_Toshiba_32G-Byte>

<LPDDR3_POP_SK Hynix_933MHz>



LG U+ SKT KT

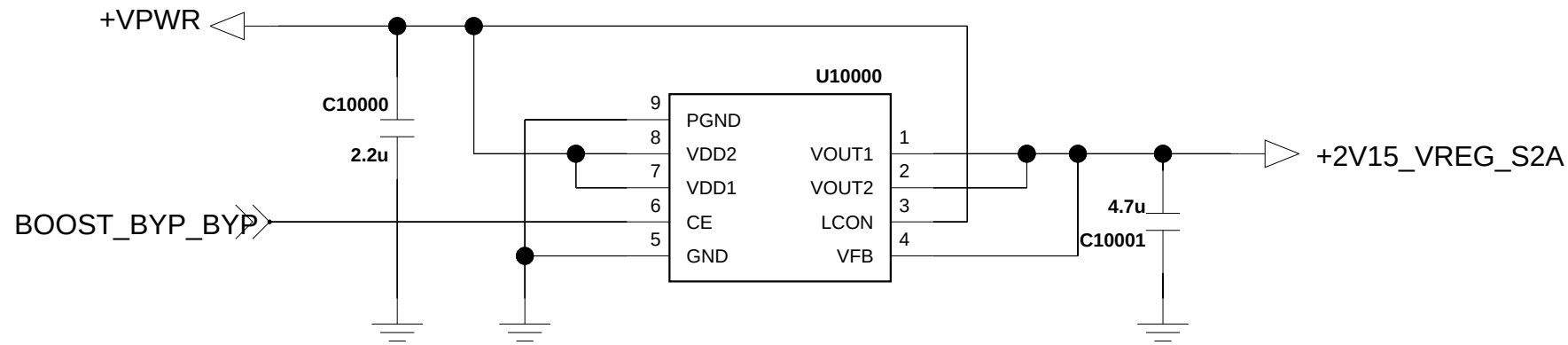


<4-1-2_PMIC_PM8941>

QCT platform uses battery FET Rds,on as sense resistor for BMS.
It can use external sense resistor if required.
If external sense resistor is not used, BMS_CSP and BMS_CSM must be tied to VBAT of PMIC.

Note : VDD_L8_L6_L8_L19 must be tied to VPWR.
The Following PMIC input pins must be tied to output of boost bypass(VPWR_BOOST_BYPASS) :
VDD_L9_L6_L17_L22, VDD_L13_L20_L23_L24, VDD_L21

Bhelper LDO



CCDS CARD Information	
Release Date	2013.08.26
Based on Reference Schematic Rev.M	
80-NA437-41, MSM8974-MSM8974A0, BASEBAND_REFERENCE_SCHEMATIC Rev.M	

0829 Using Rds : R15113,R15114(DNI), R15138, R15139(0ohm)
USB IN lines of the PMIC for noise rejection and for charger removal detection.
It's near by USB connector

Rev 0.3

If DC IN path is not used, short DC IN_OVP_SNS to DC IN and tie leave DC_IN_OVP_CTL floating.

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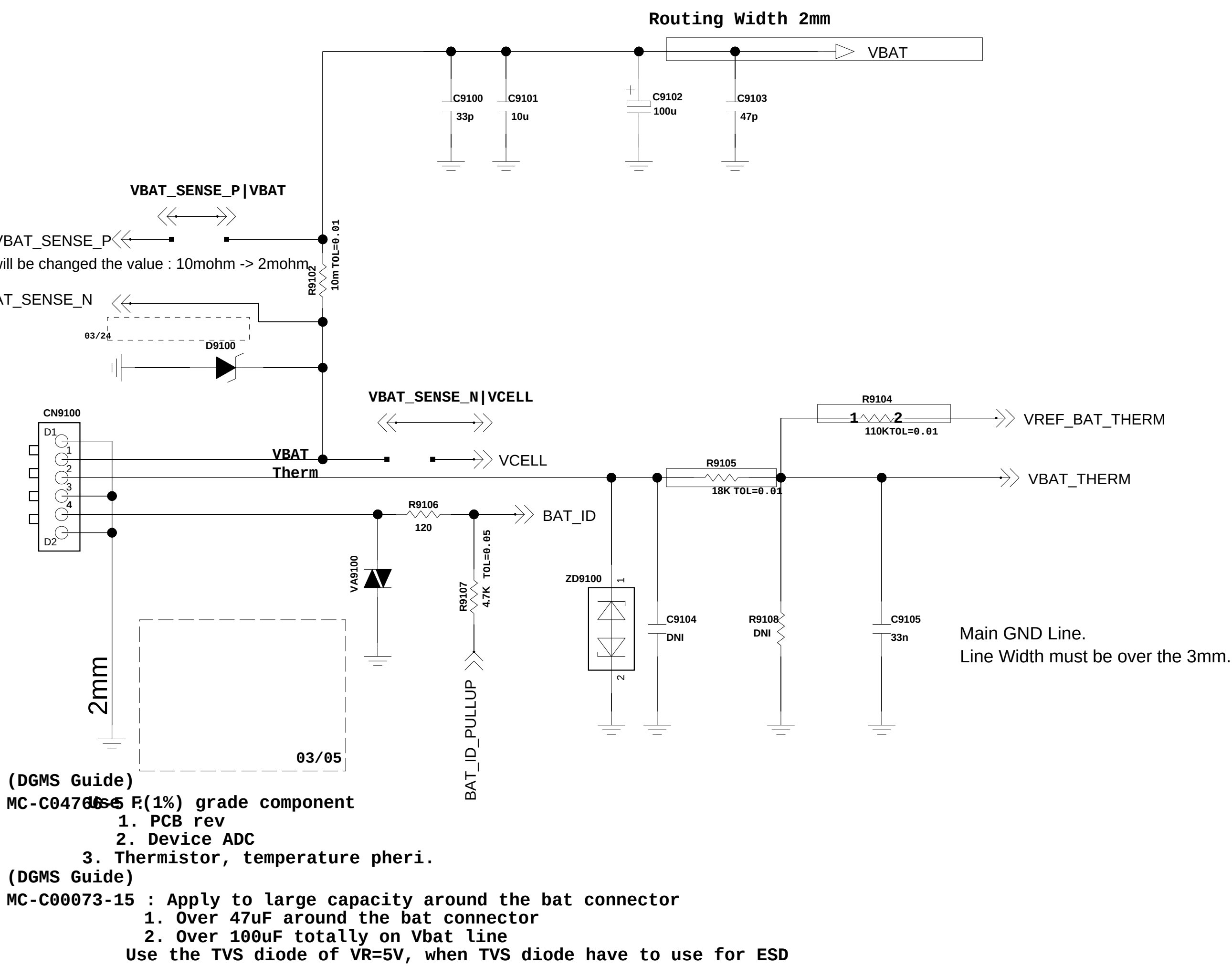
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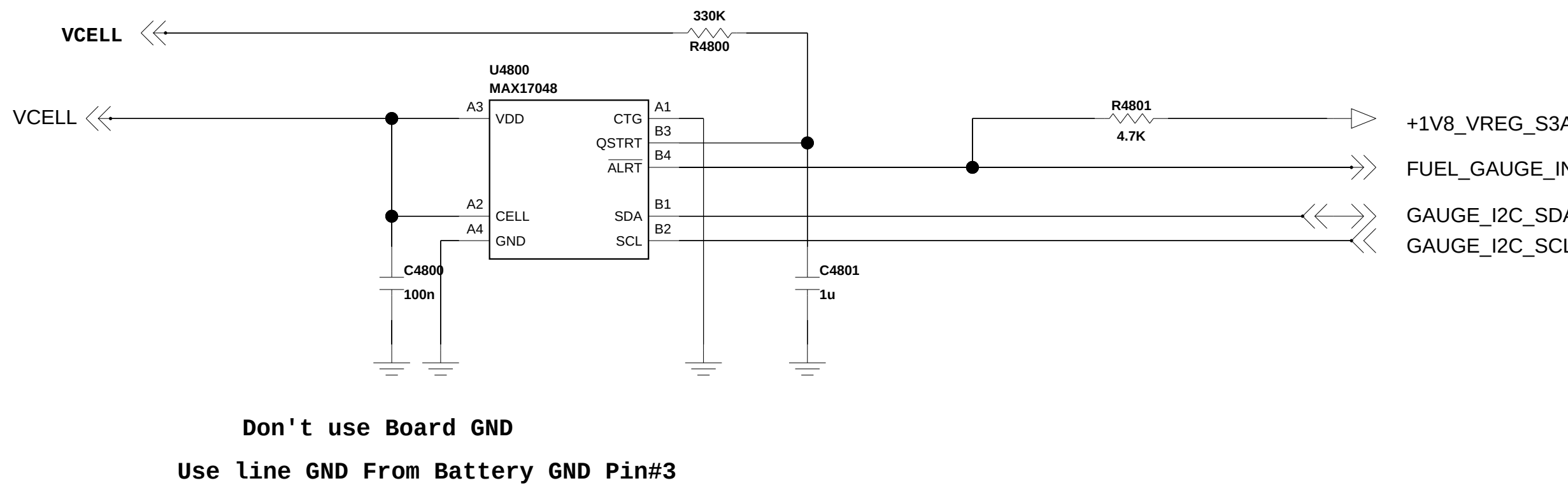
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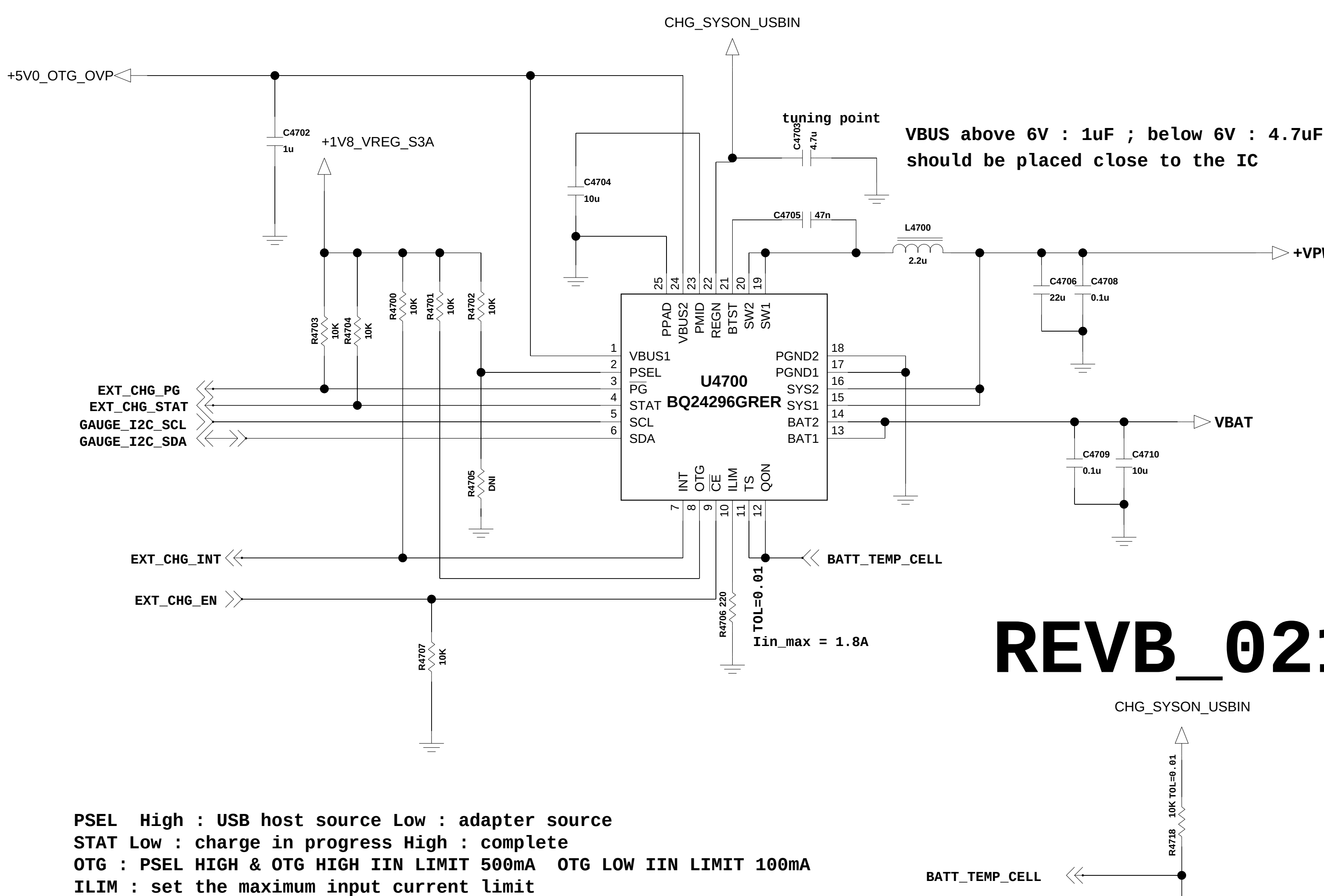
<4-8-1_Fuel_Gauge_MAX17048>Rev_0.3

**Only use line VBAT1 from Battery Connector
CLOSE TO Battery CONNECTOR**



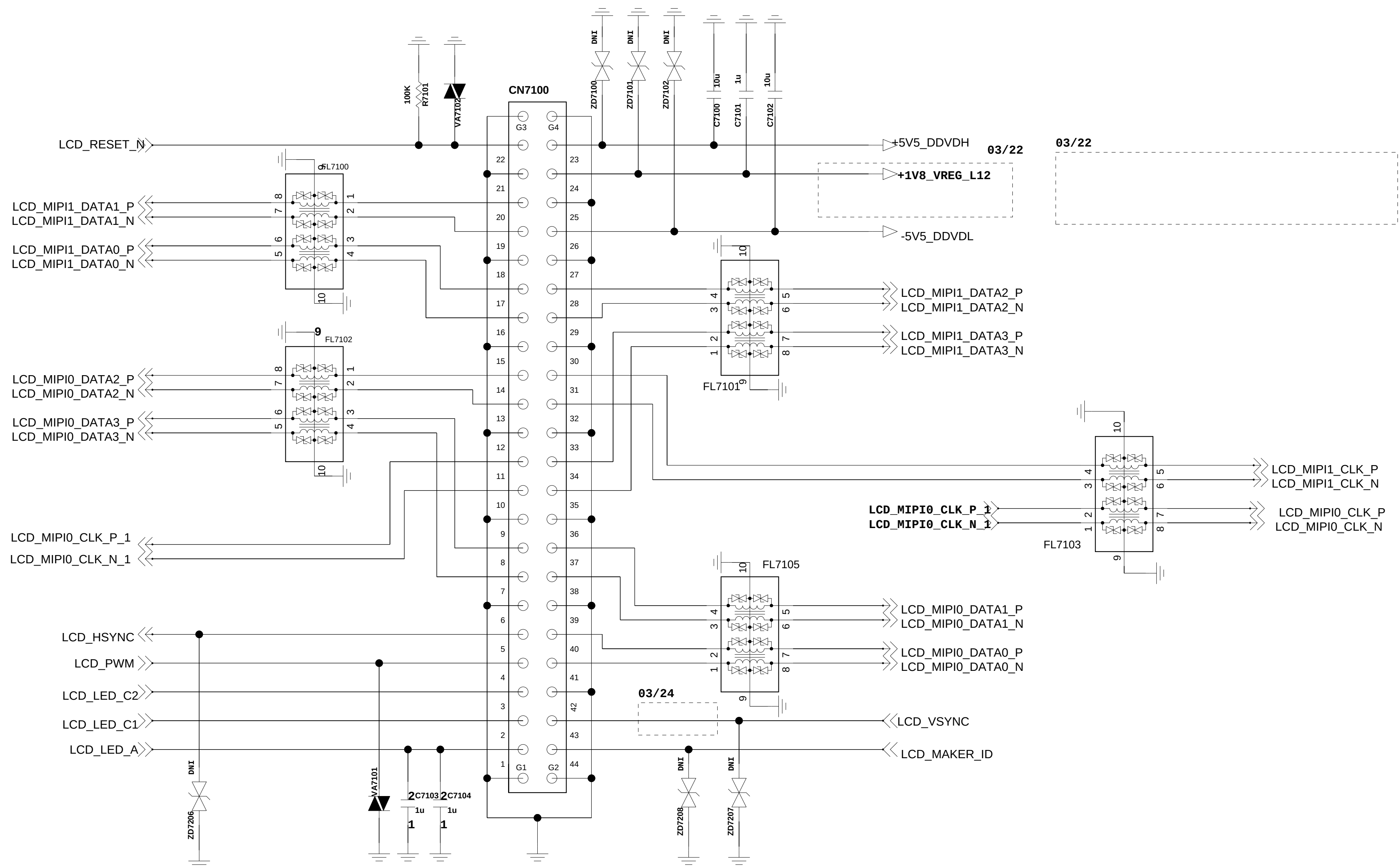
Single charger IC

<4-7-2-4_Charger_Switching_BQ24192>

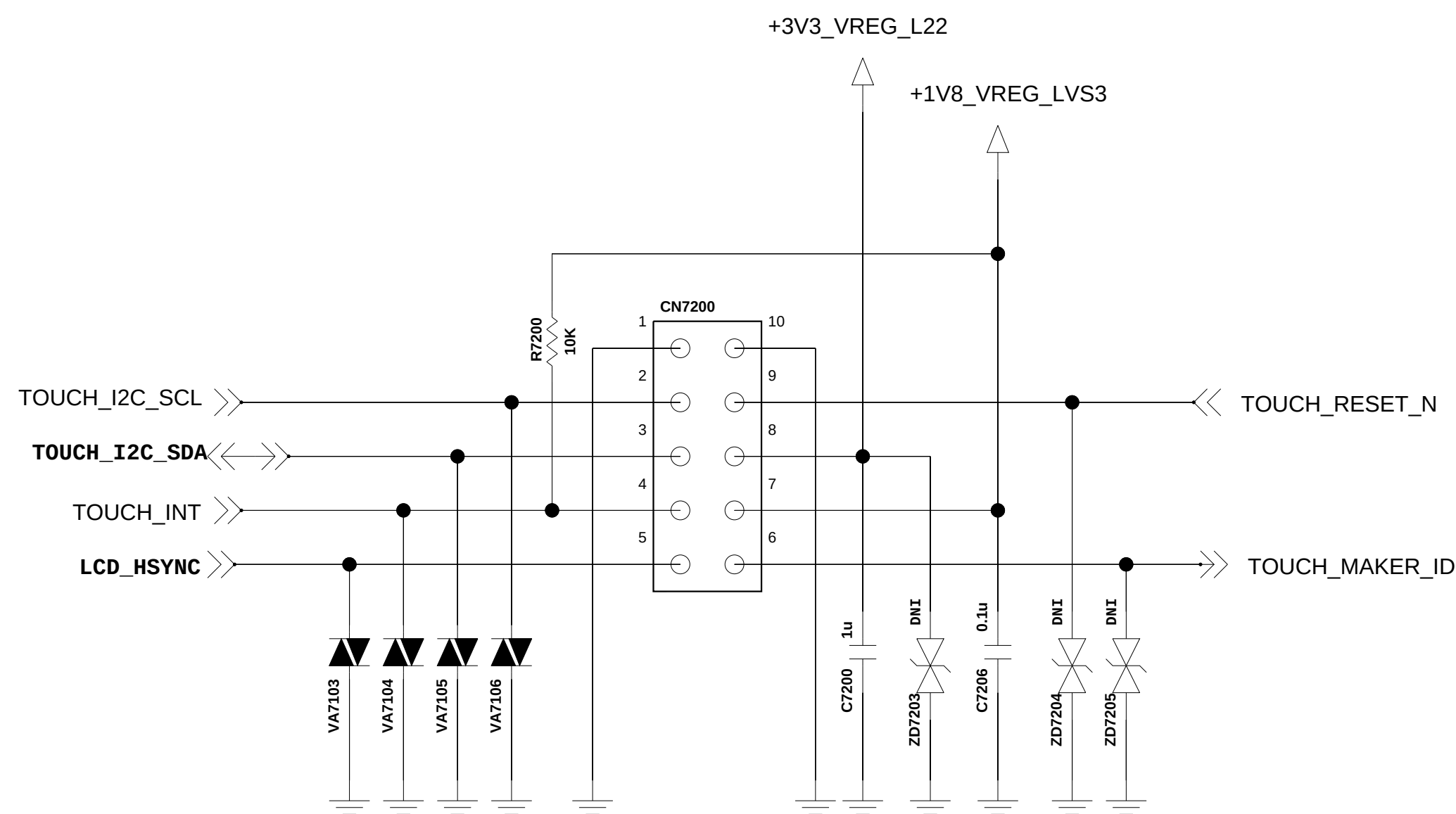


REVB_0213

<7-1-8-4-2_Quad_LCD_5.5">Rev_0.1



5.5" Touch

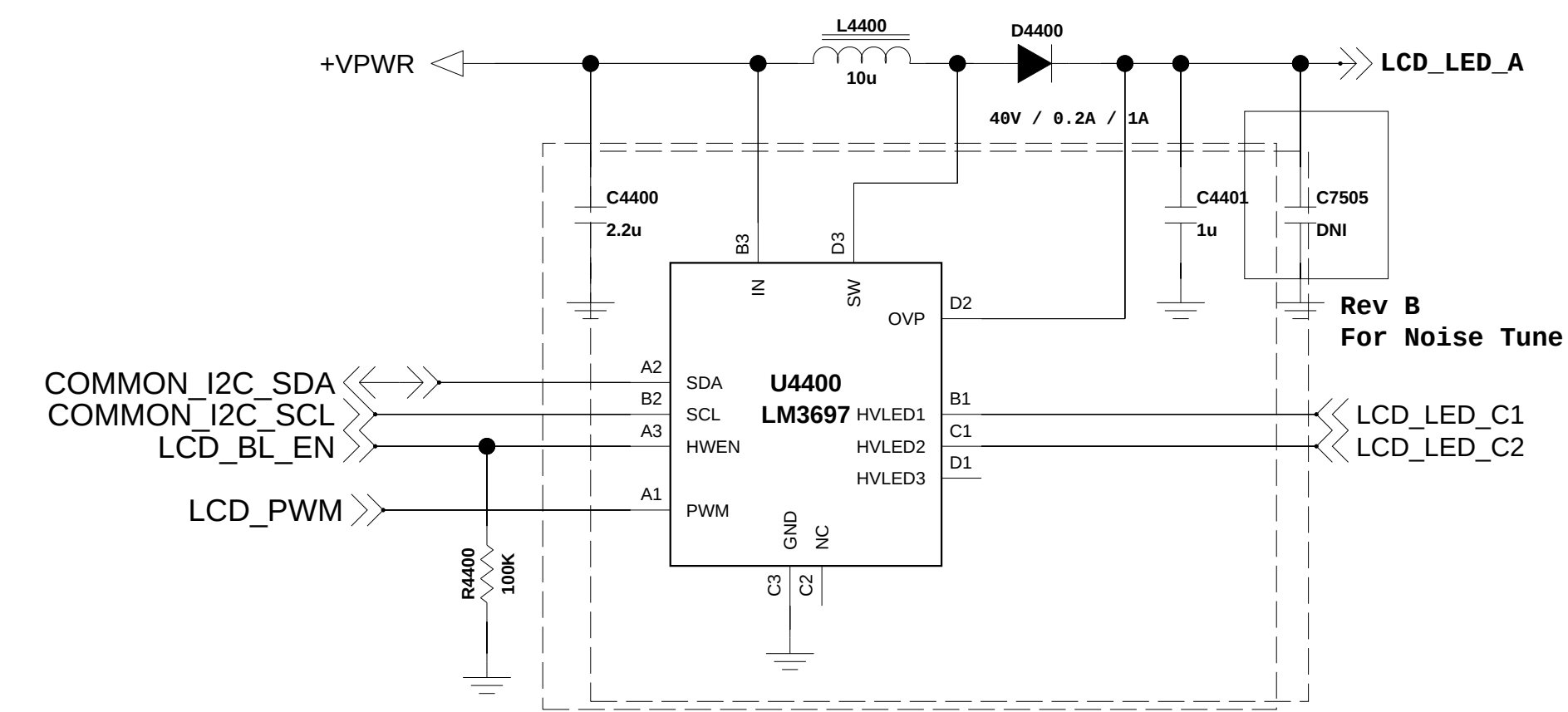


03/15

<4-4-4_LCD_BL_Booster_LM3697>

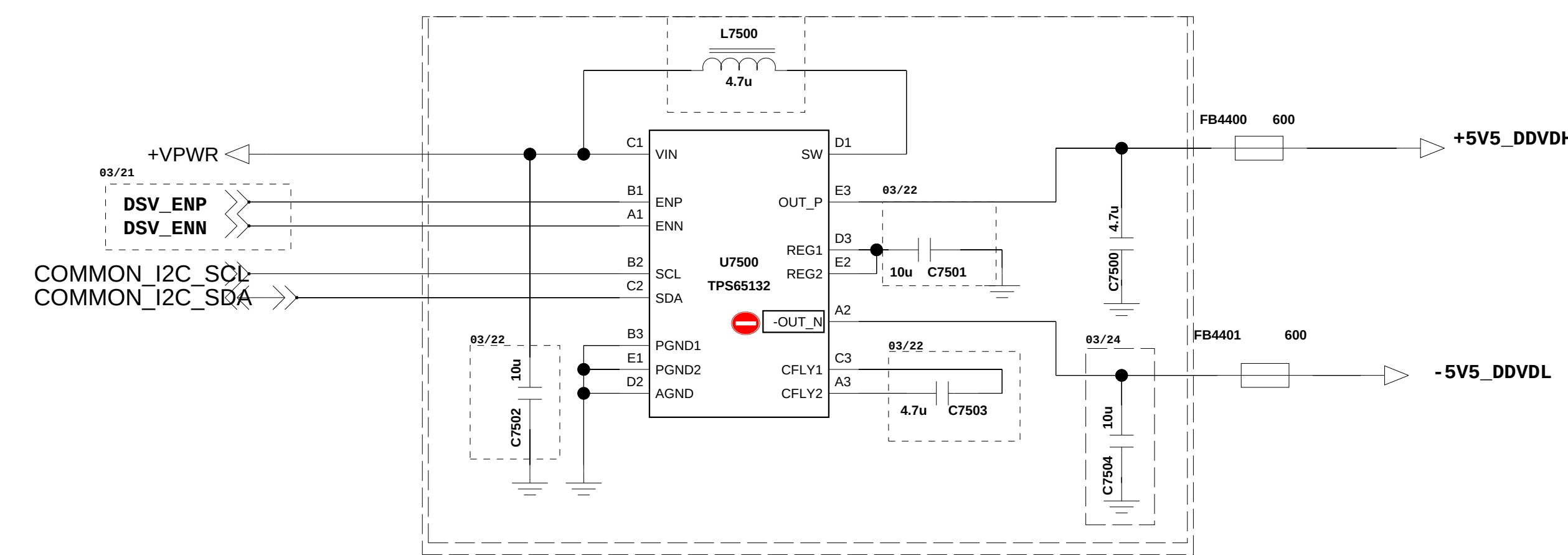
Rev_0.3 Smartphone mode

LED Vf=3.2V, If=23mA/46mA(2string), 2P7S
20% inductor tolerance => fsw=500kHz : Ipeak = 722.6mA, fsw=1MHz : Ipeak = 541.9mA
30% inductor tolerance => fsw=500kHz : Ipeak = 774.2mA, fsw=1MHz : Ipeak = 567.7mA
LED Vf=3.0V, If=20mA/46mA(2string), 2P7S
20% inductor tolerance => fsw=500kHz : Ipeak = 651.9mA, fsw=1MHz : Ipeak = 473.3mA
30% inductor tolerance => fsw=500kHz : Ipeak = 702.9mA, fsw=1MHz : Ipeak = 498.9mA
If you have other LED spec&quantity, recalculate Ipeak And select Power inductor
Vout_max = 22.7V at Vf=3.2V, 2P7S, Headroom V=0.3V
You select diode spec : VR > Vout_max, If > LED If(all string current), I_FRM > Inductor Ipeak



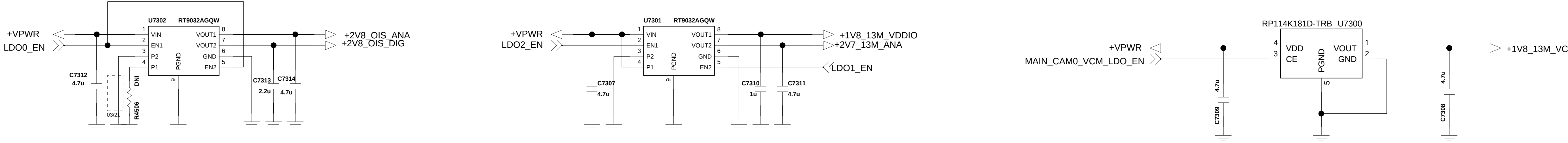
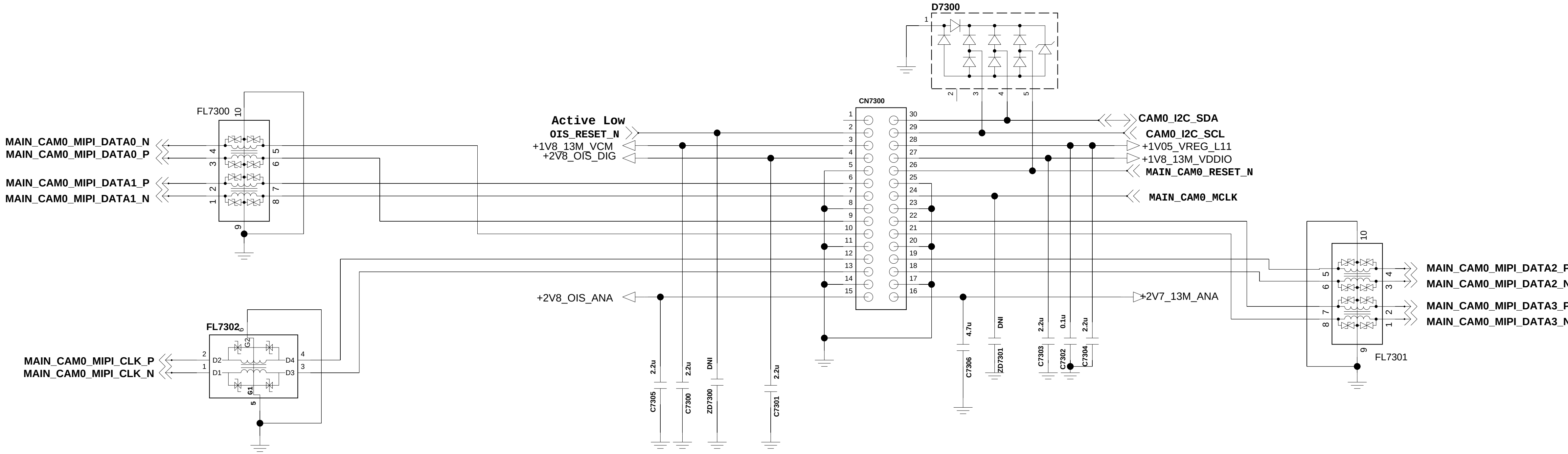
03/22

<7-5-1_DSV_TPS65132>Rev_1.0



DGMS Guide
MC-C06162-1 : 1. Don't connect Positive of Tantal, TVS diode to Negative voltage if use
2. We recommend MLCC capacitor in this case

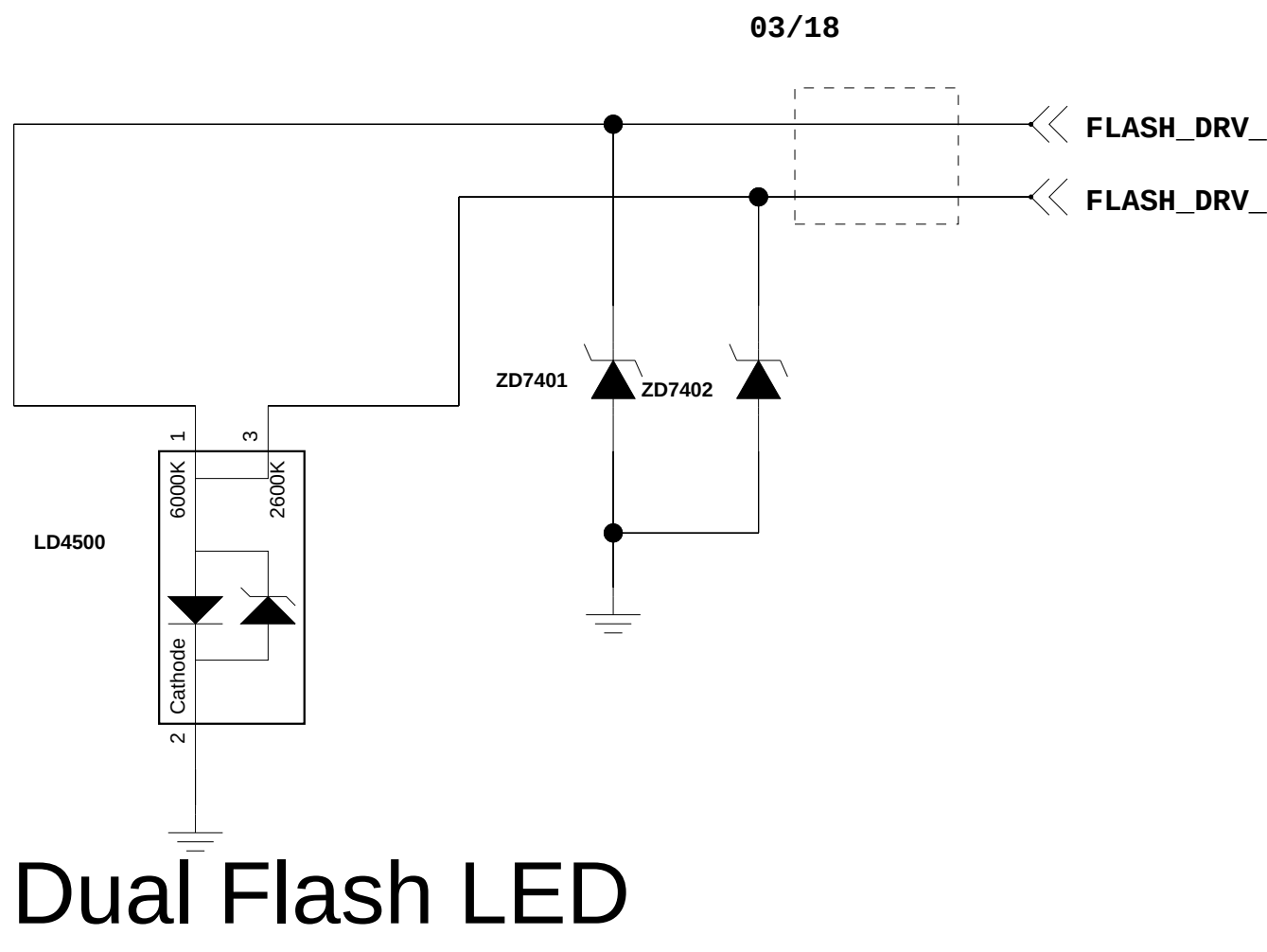
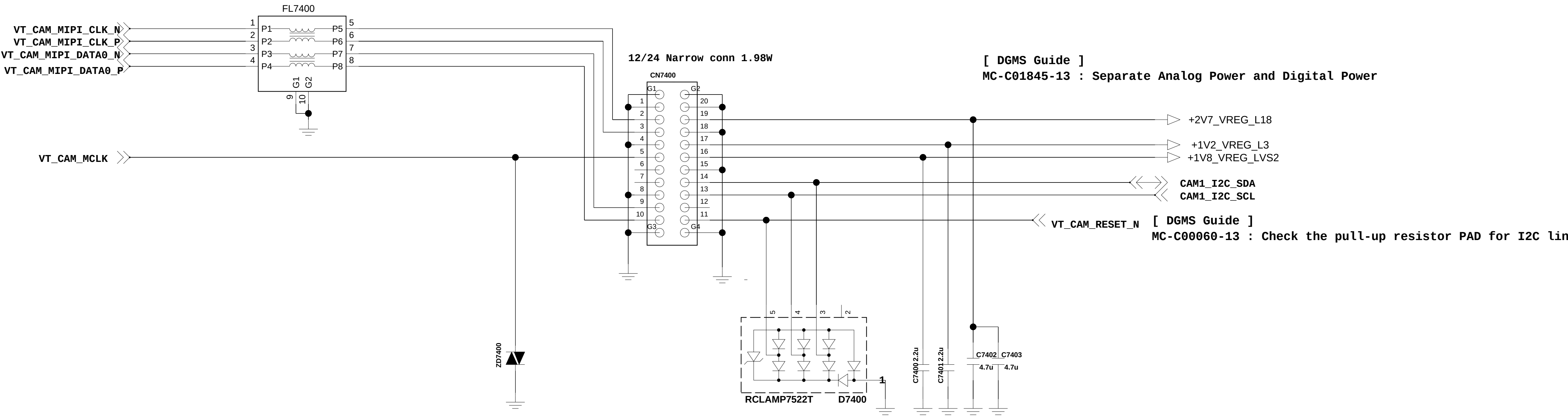
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< 7-4-1_VT_Camera_20pin_VGA_1.26M_1.3M_2.4M>

[DGMS Guide]

MC-C00081-17: 1. Add ESD Protection circuit for Camera signal lines
2. Use under 100ohm/15pF ESD protection for over 2MP Camera



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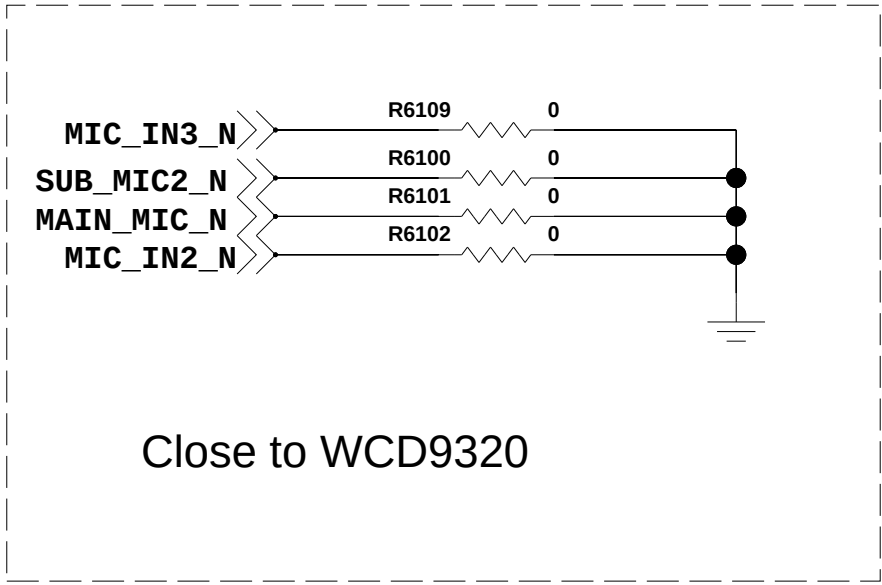
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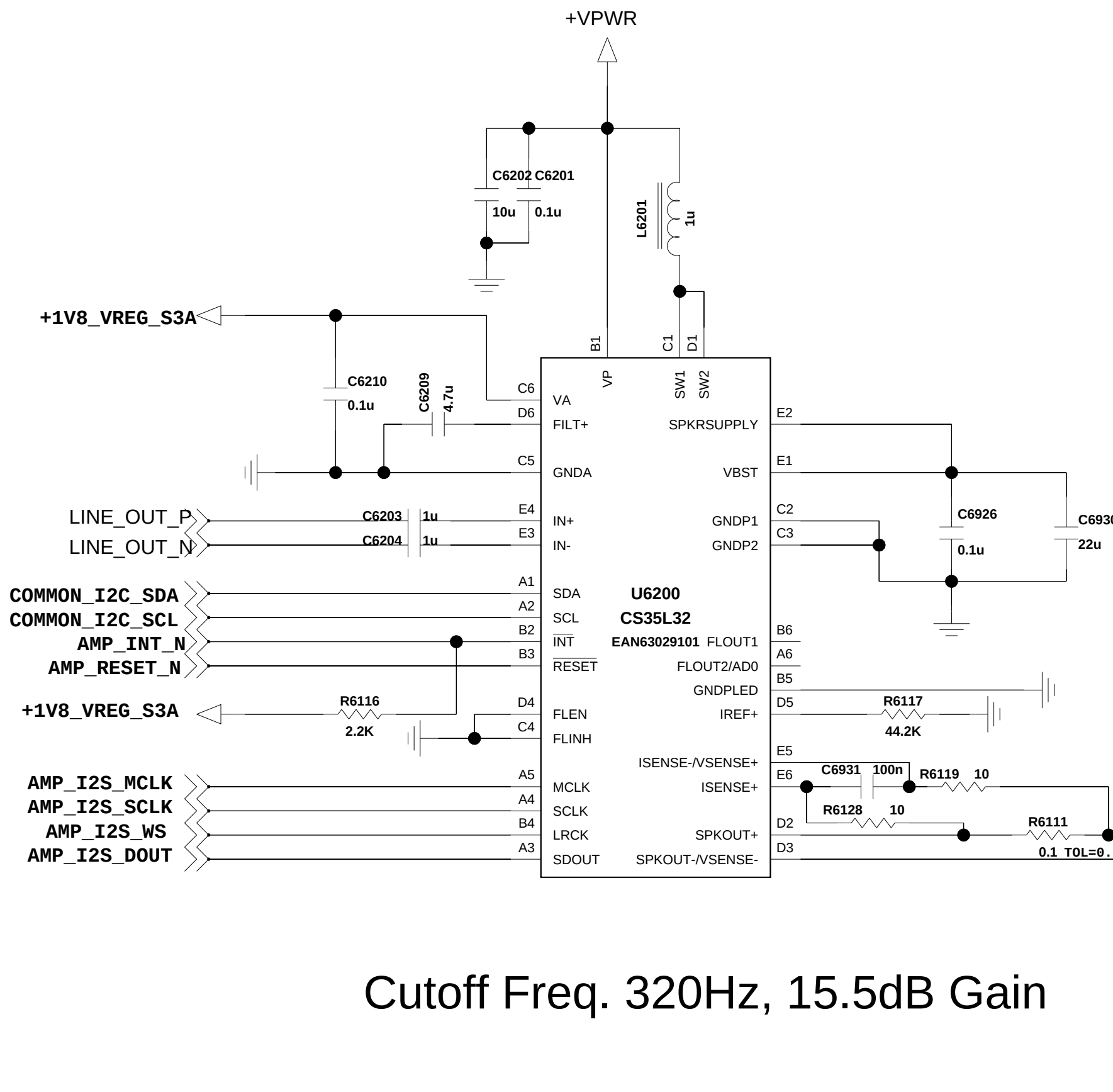
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Close to WCD9320

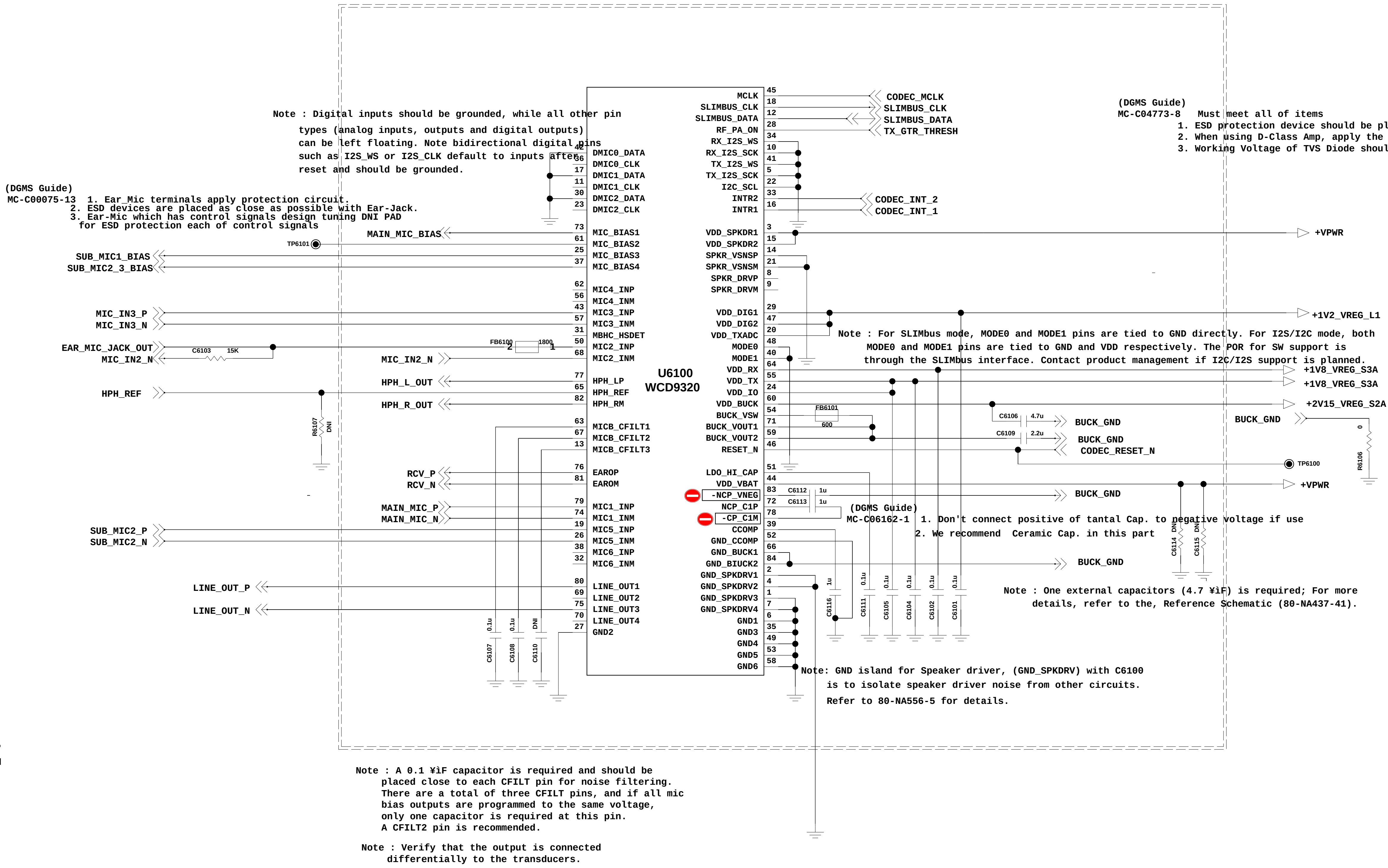
SPK BOOST AMP



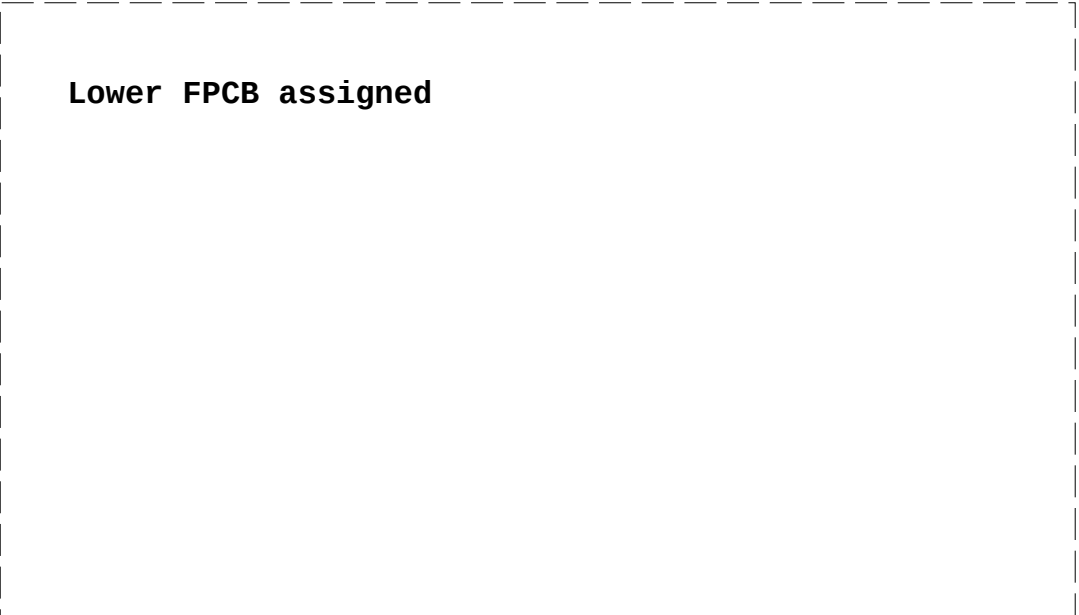
Cutoff Freq. 320Hz, 15.5dB Gain

<6-1-1-5_WCD9320>

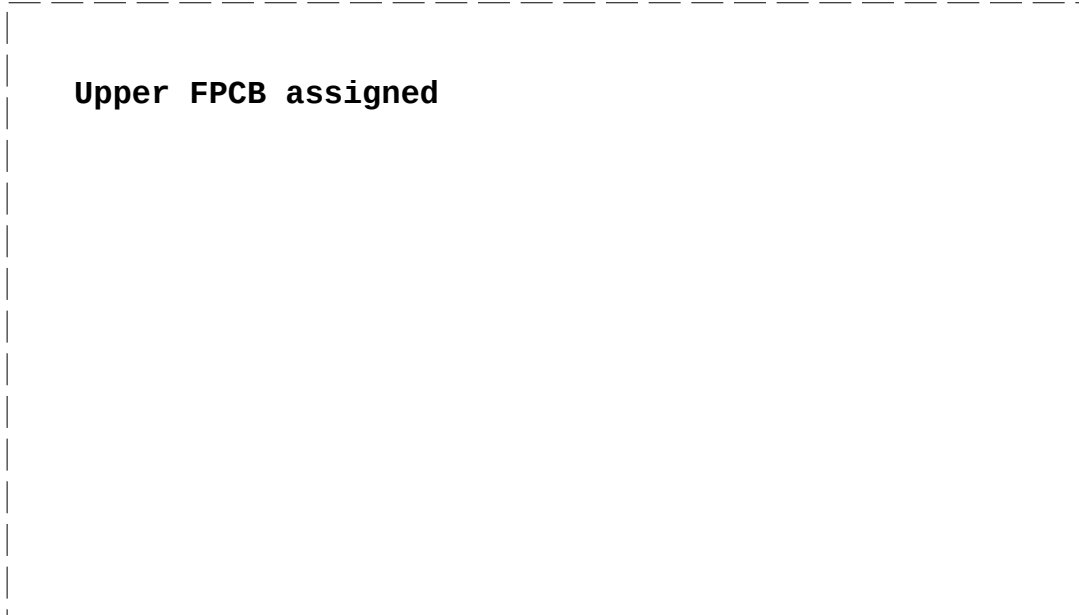
REV_0.3



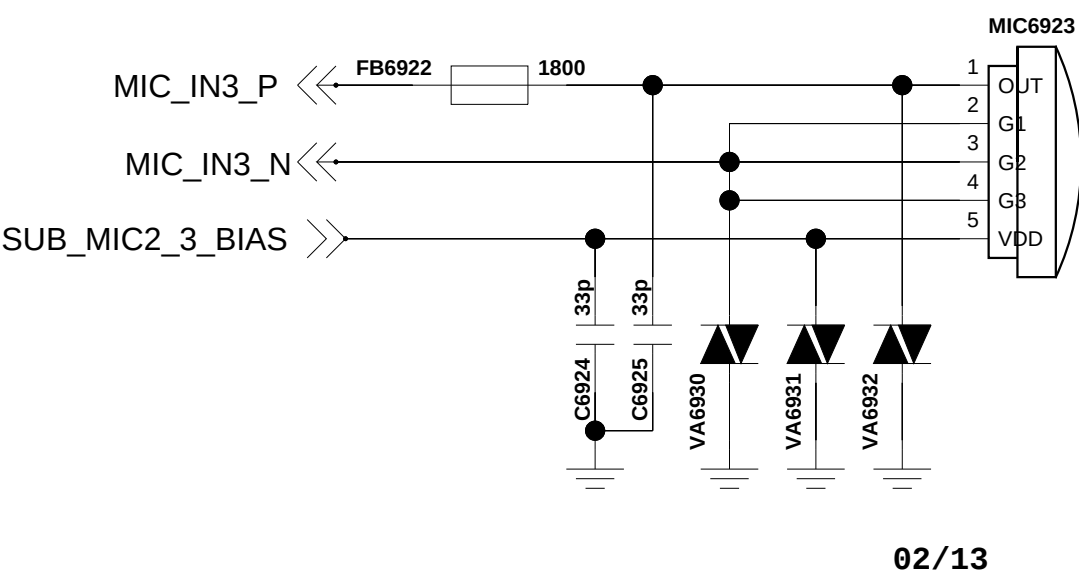
Main MIC



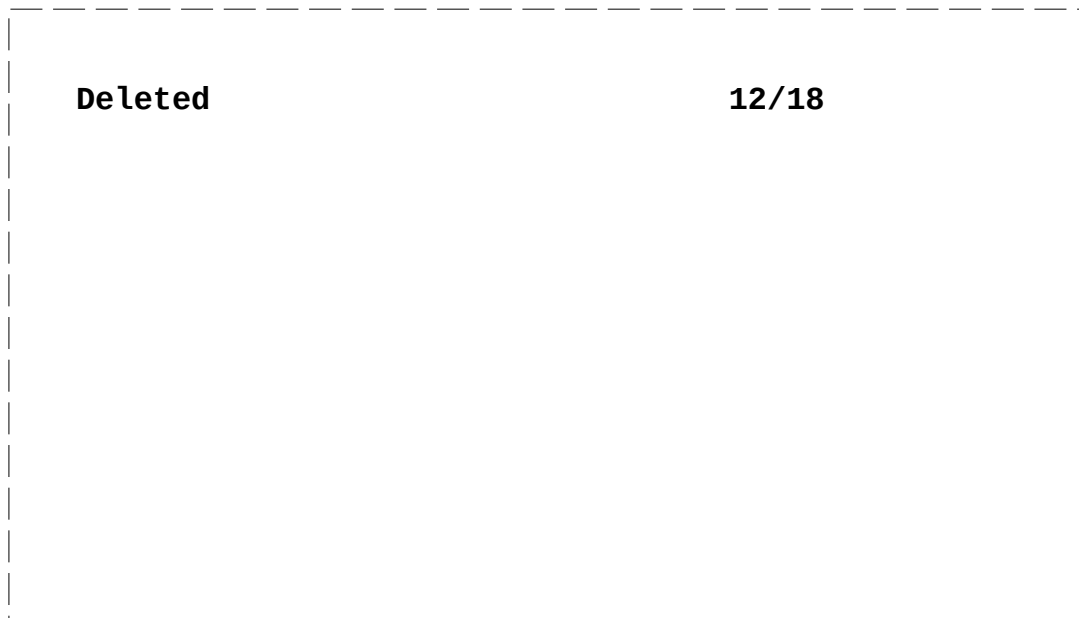
Sub_MIC1



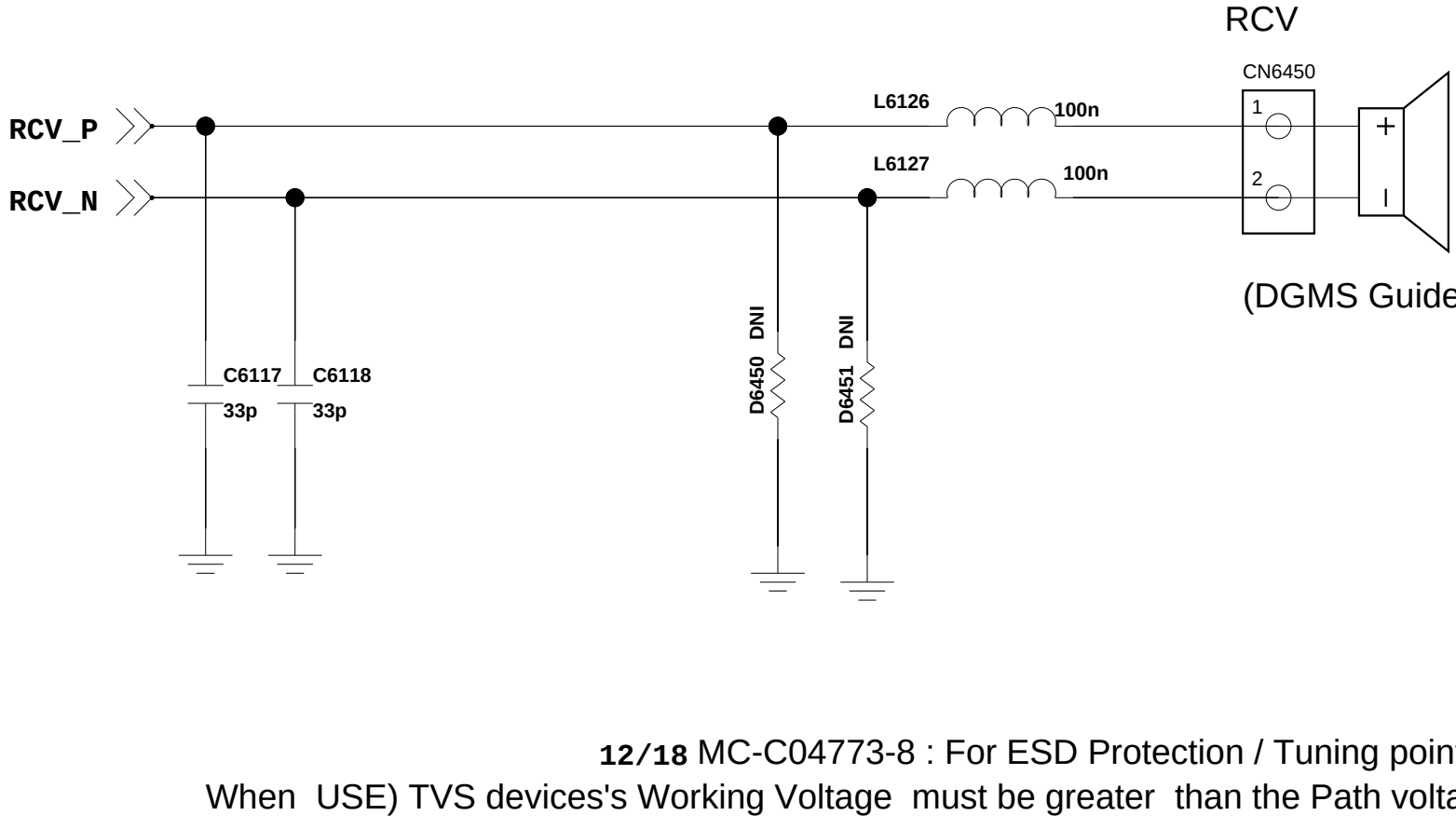
Sub_MIC2



Sub_MIC3

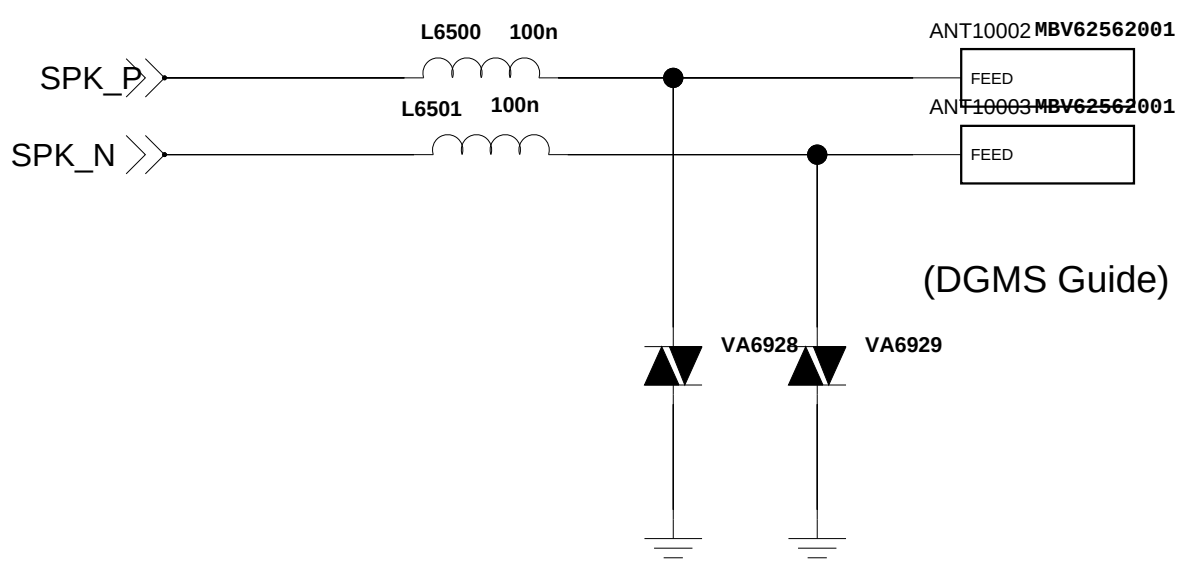


< 6-8-2_Receiver >



< 6-7-1_Speaker >

Rev_0.4



LGE Internal Use Only

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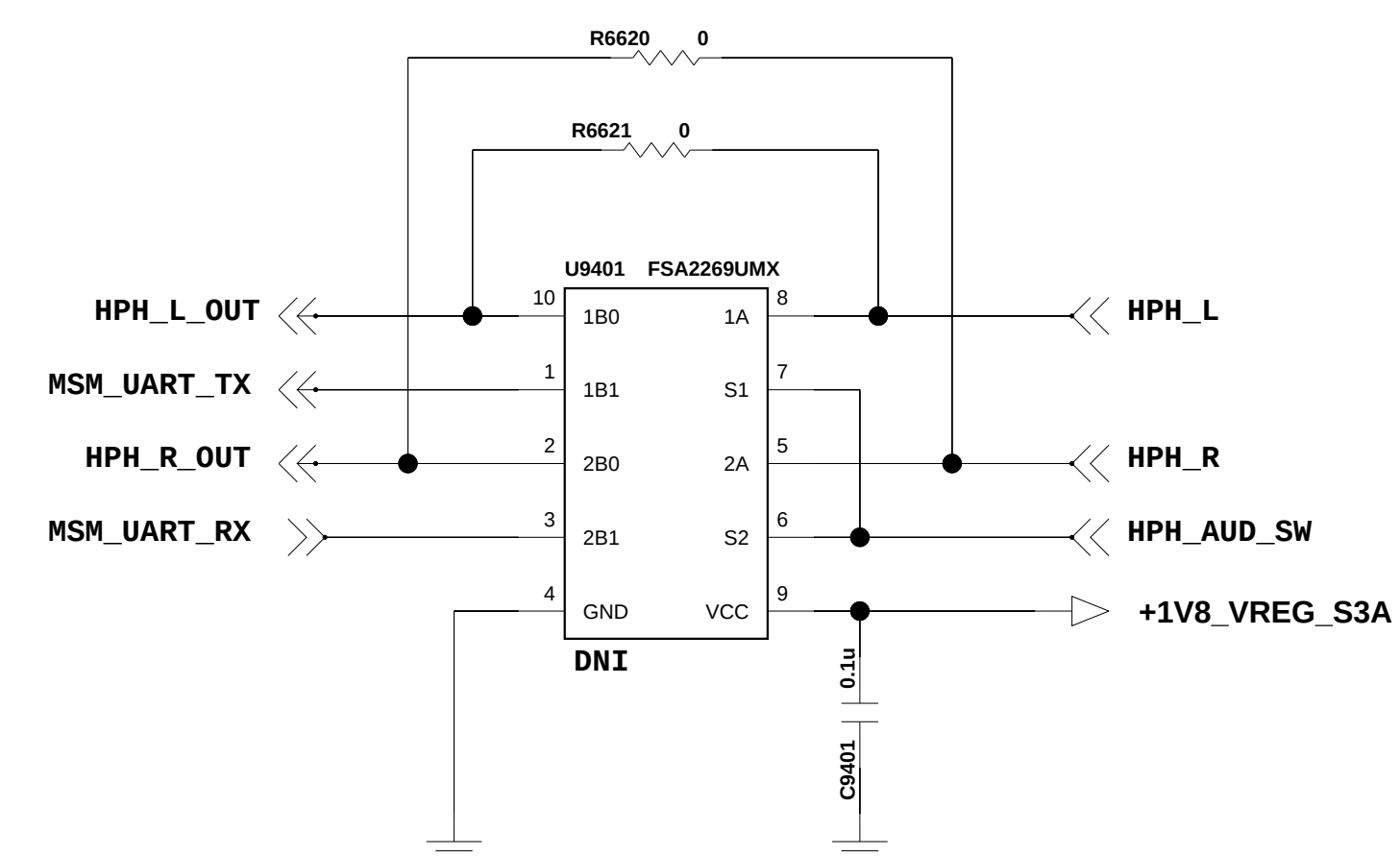
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<Earjack_Debug_Port >



(DGMS Guide)
MC-C00075-13 : For ESD protection / tuning point

Rev_0.4

EAR_SENSE_JACK <<< F86611 1000 F86612 1000 F86610 1000

HPF L
HPF_R

EAR_MIC_JACK <<< F86610 1000

HPF_REF >>>

GND Close to EarJack

LB100 100n SW_Detect 36610

LB101 100n SPK_L

LB102 100n SPK_R

LB103 100n GND

LB103 100n MIC

C86616 22p

C86618 22p

C86615 22p

D86600

D86601

D86602

D86603

D86604

R86611 0

Place Close to the HPF Ref. w. Thru Via to L

Place Close to the EJ Conn.
w. Thru Via to Large GND

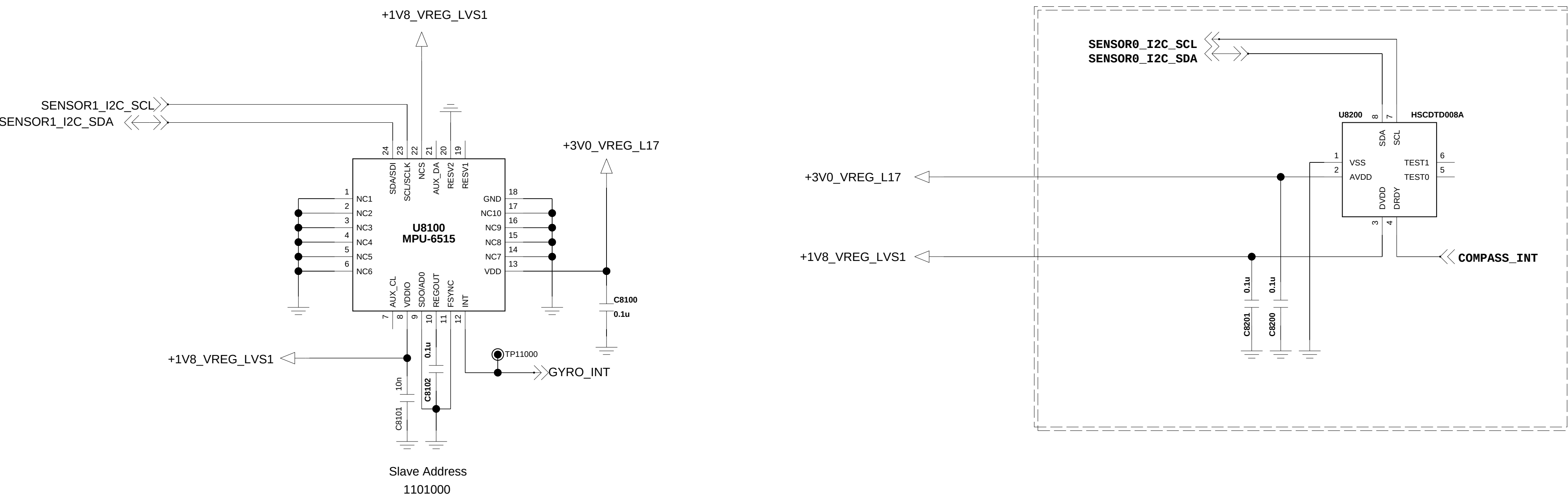
SENSOR/MOTOR/IRRRC/SIM/SD/LED

I2C	Devices
Sensor0	Compass, Temp+Humid+Pressure
Sensor1	Accel+Gyro, Pressure, UV, Prox+RGB+IR Gesture

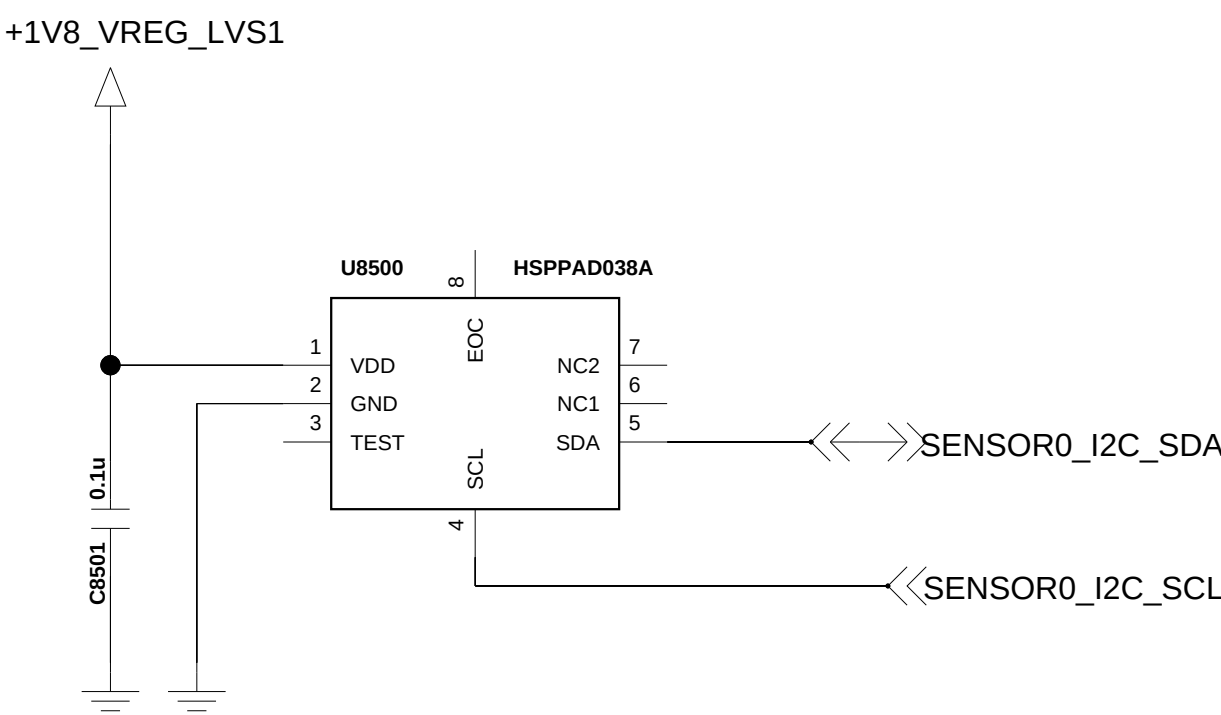
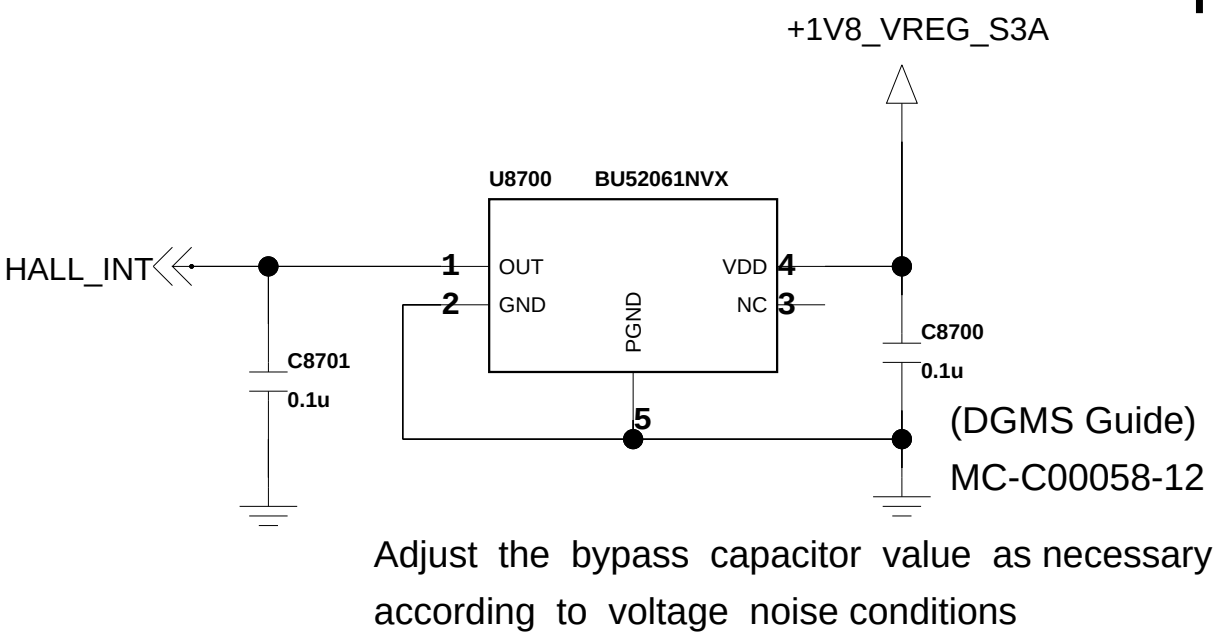
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< 8-2-1-2_Compass_HSCDTD008A>
Rev_0.3

<Pressure_HSPPAD038A >



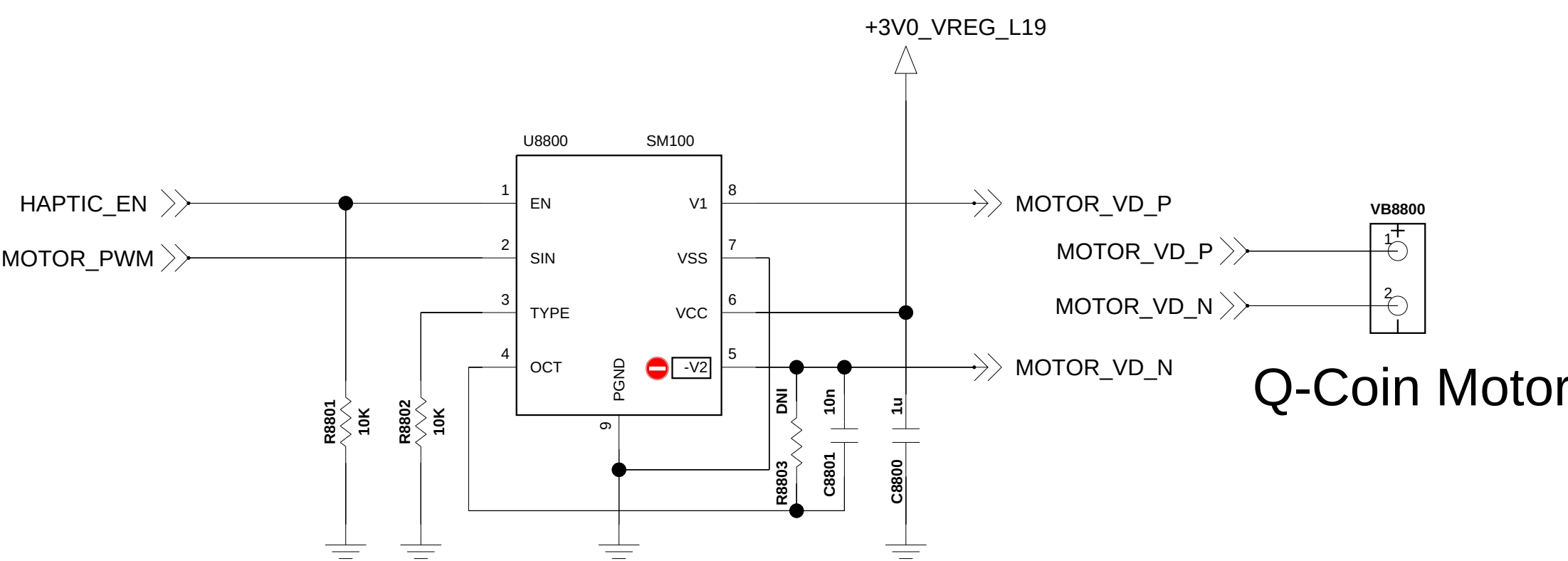
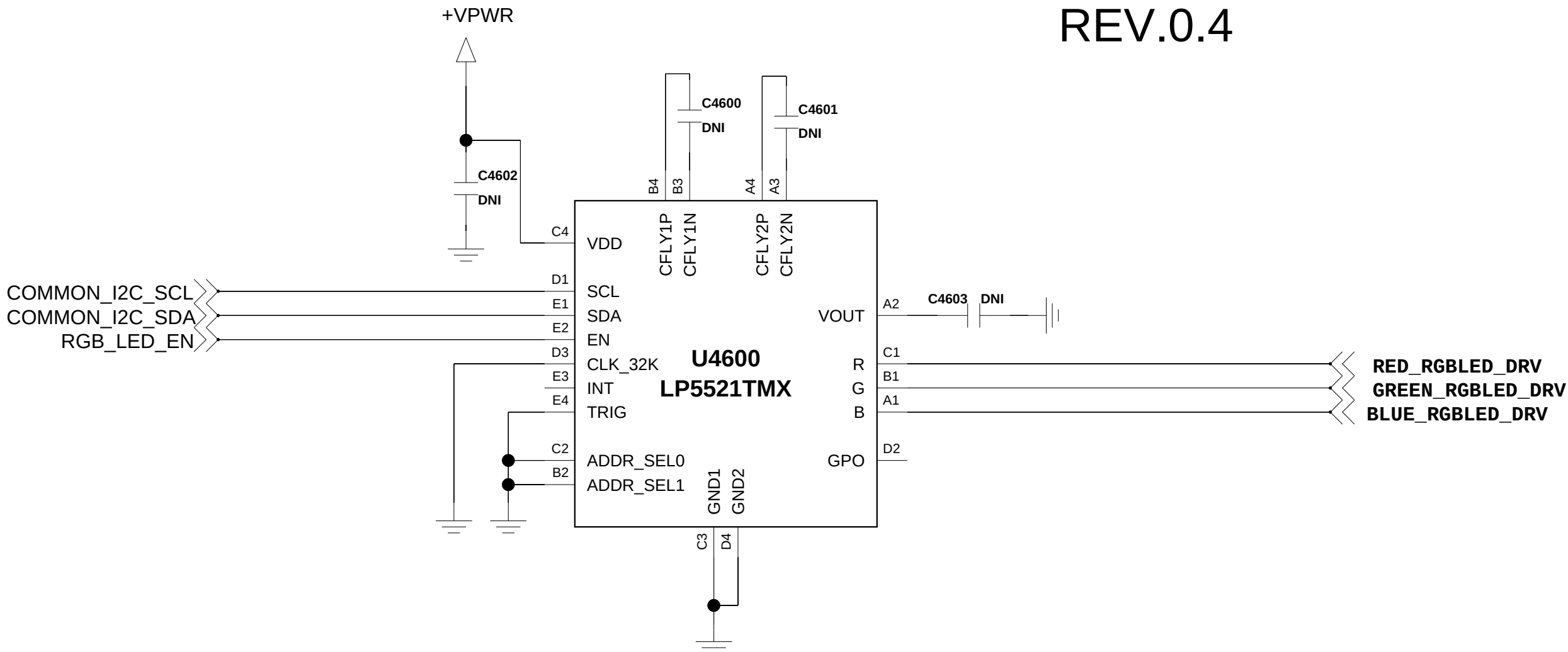
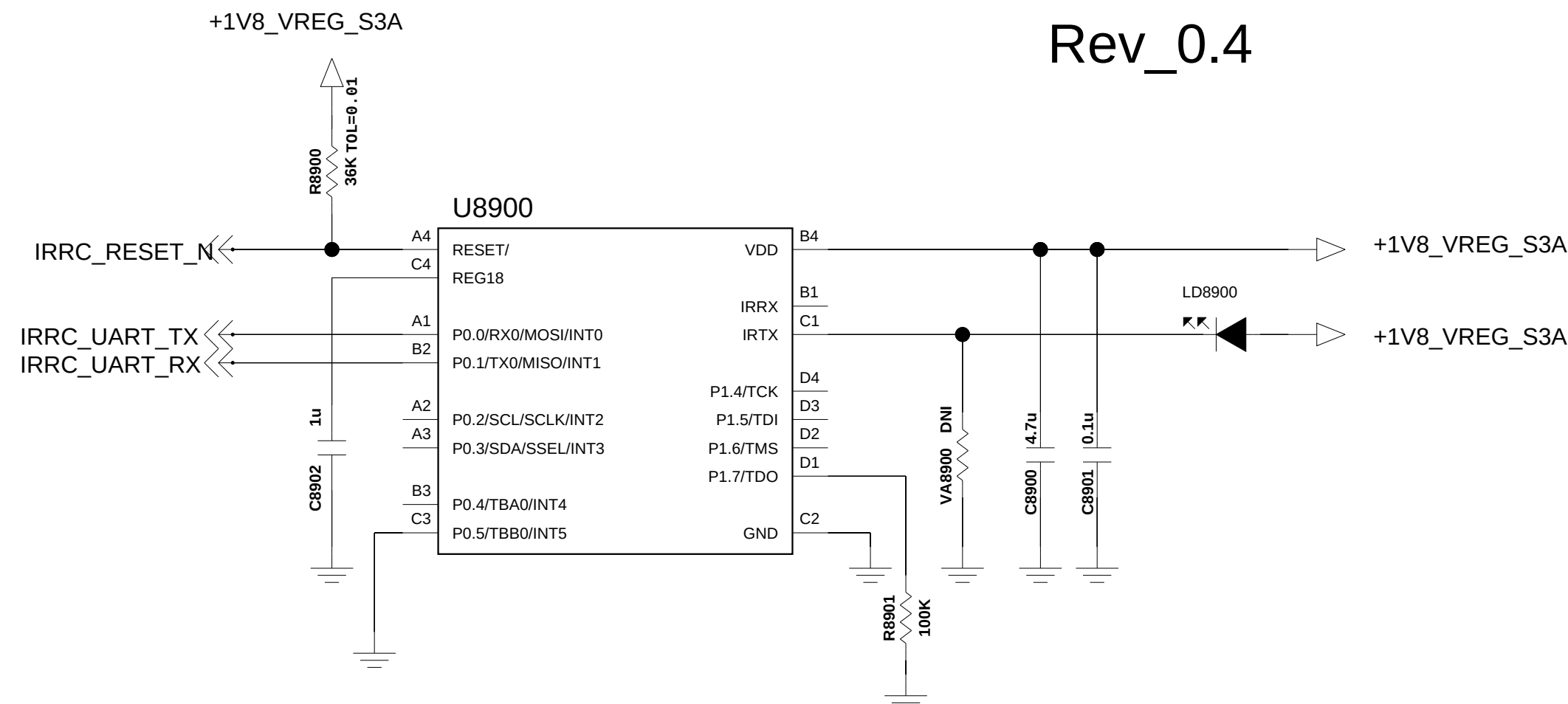
<8-7-1-1_Hall IC_BU52061NVX>
Rev_0.3



< 8-9-1-1_IRRC_MAXQ616V >

<4-6-1_LED_Driver_IC_LP5521>

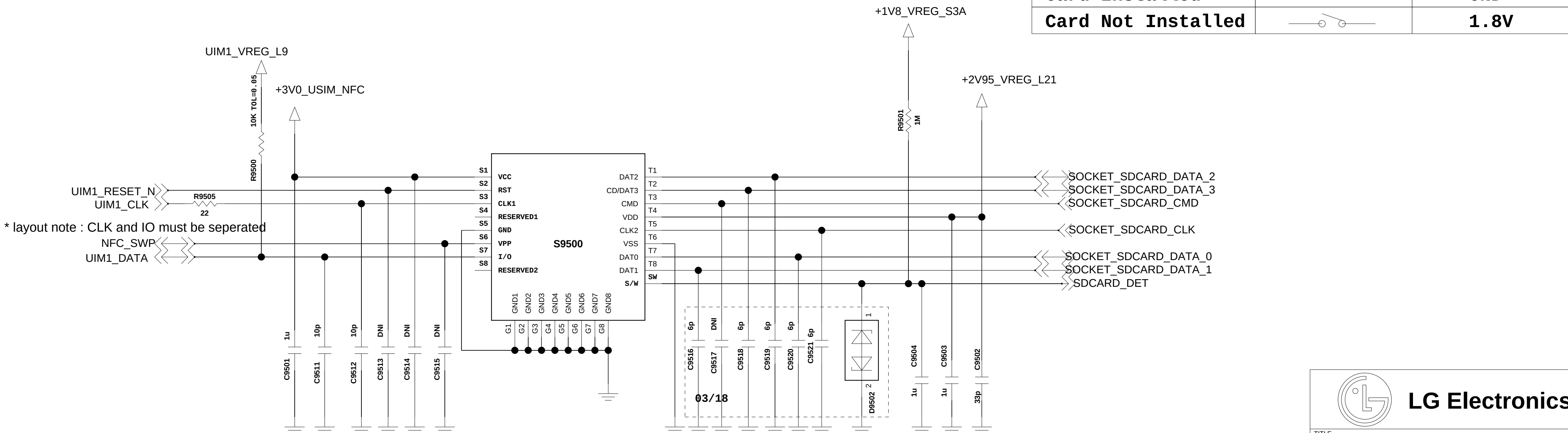
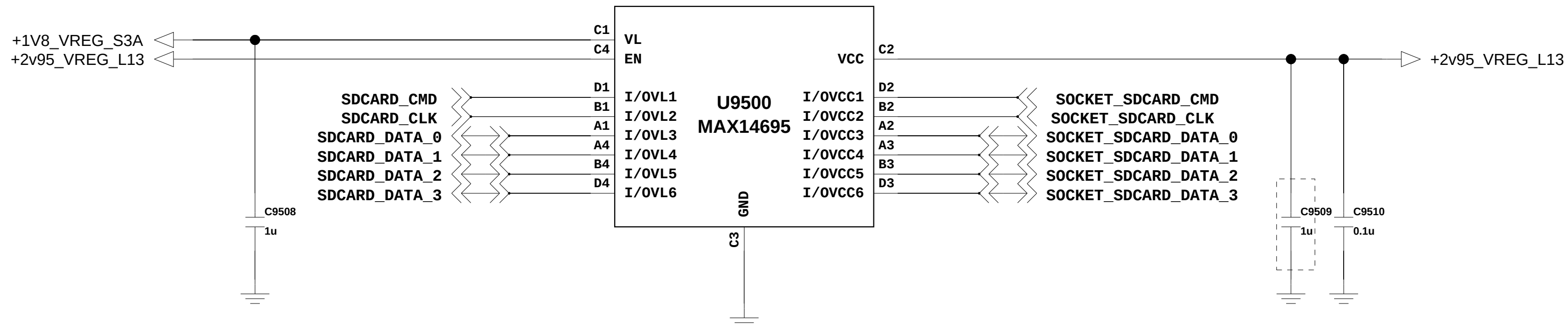
< 8-8-1-1_Haptic_SM100 >



<LEVEL TRANSLATOR>

< 9-5-3_Micro_SD_Combo >
Rev_0.3

Card Detect	SW	VSS(3V)	CARD_DET
Card Installed			GND
Card Not Installed			1.8V



A

A

A

A

A

A



L

K



1



G



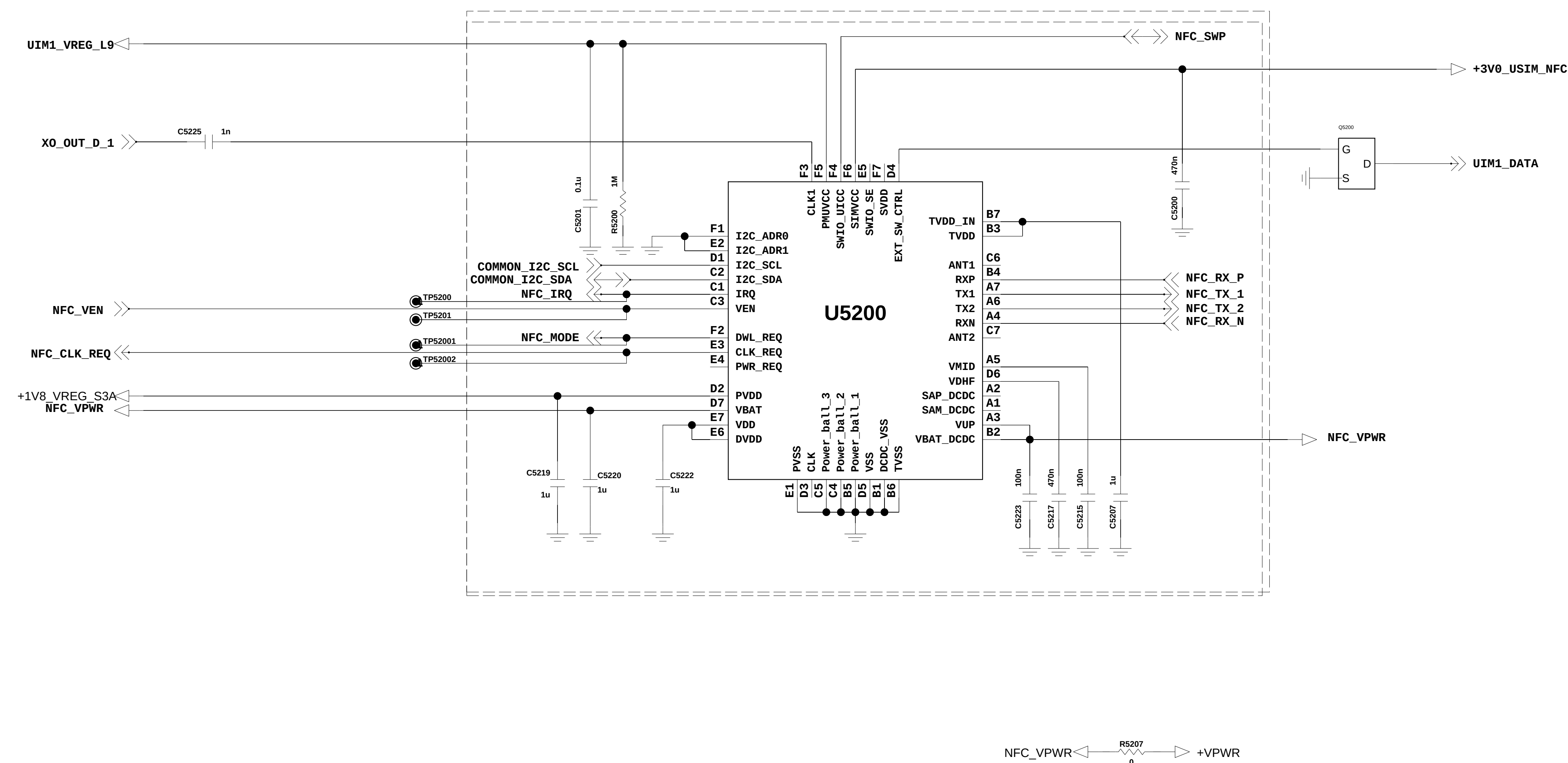
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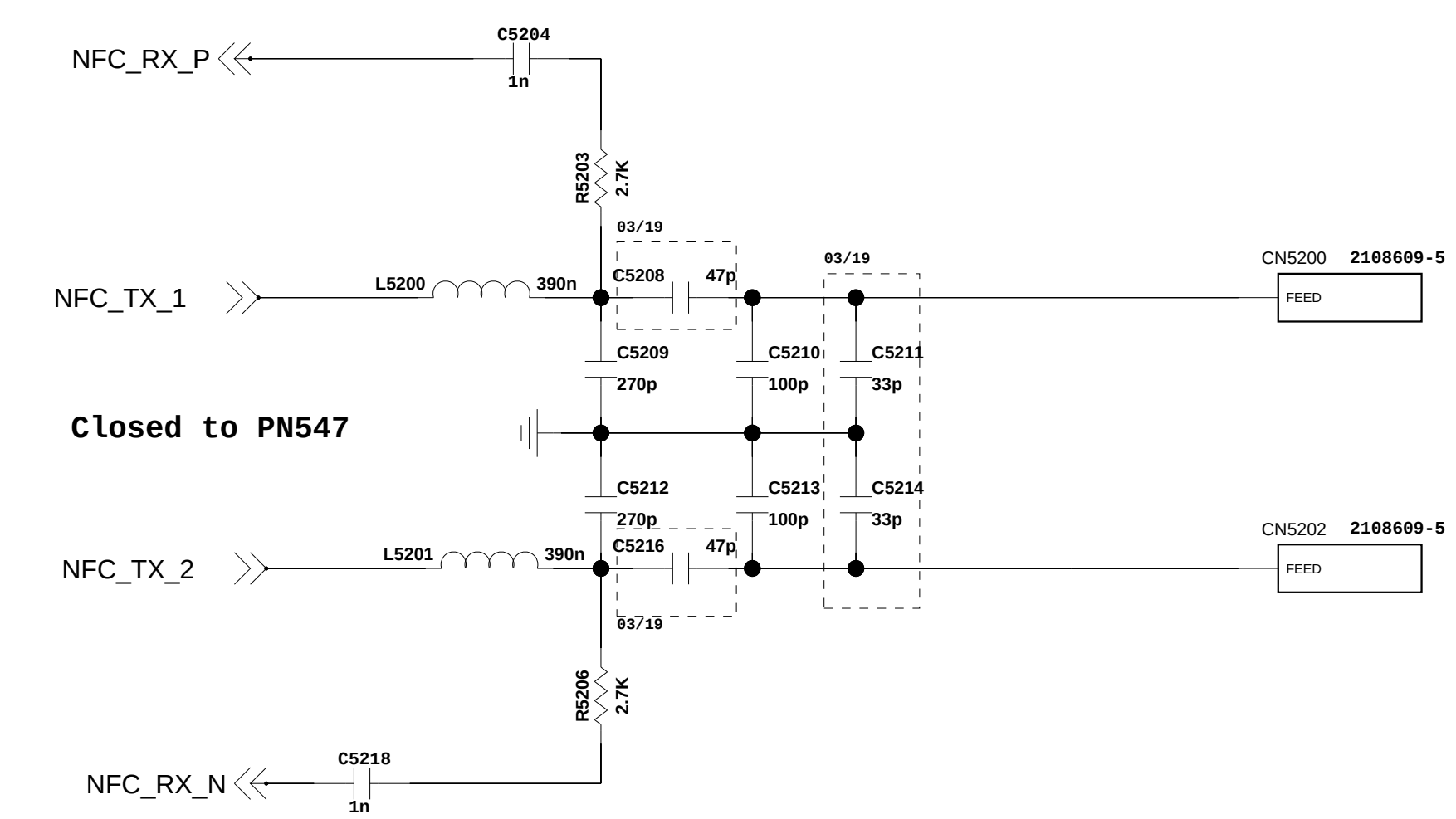
D



<5-2-1-4_NFC_PN547> Rev_0.5

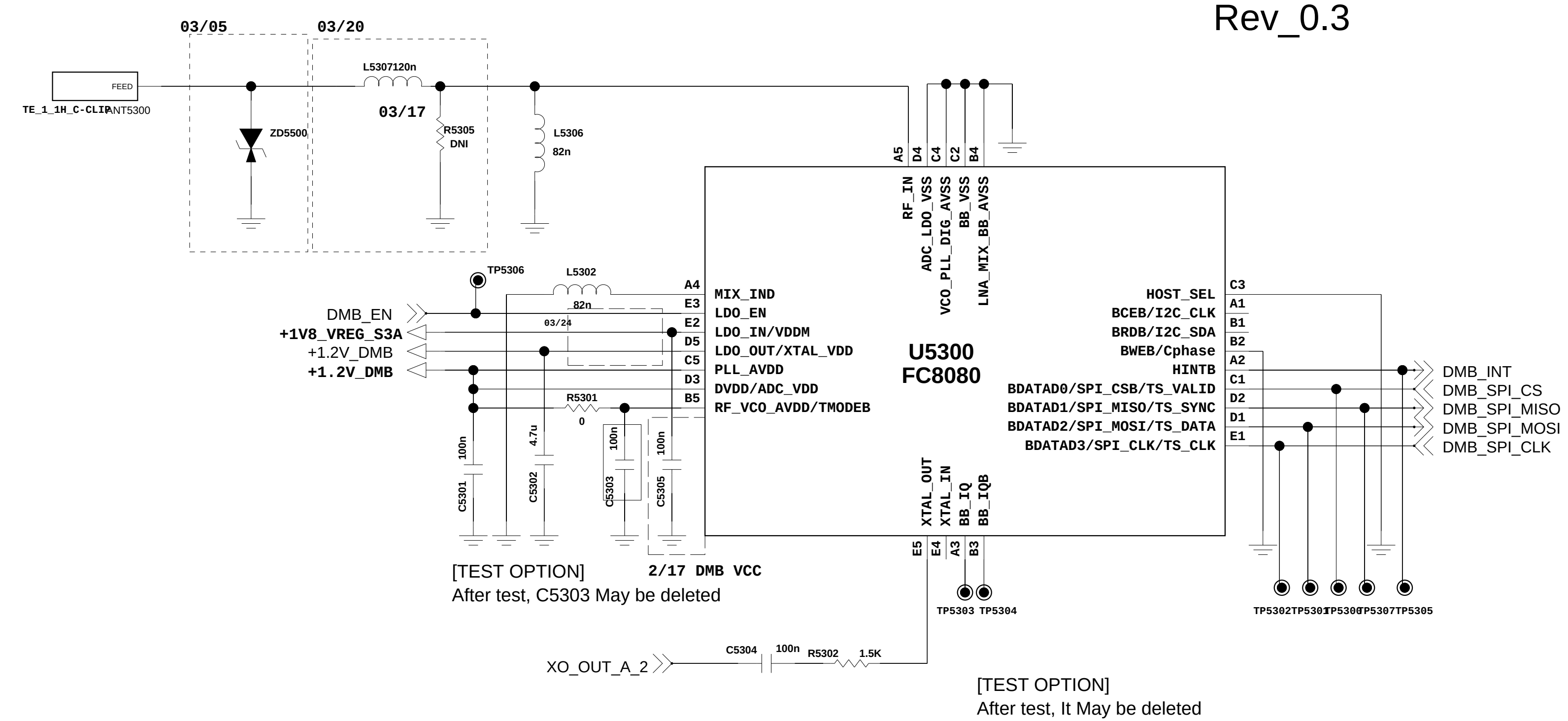


NFC Antenna



< 5-3-1-5_TDMB_FC8080 >

Rev_0.3



LGE Internal Use Only

L

K

J



H

H

G

G

F**F**

E

E

D

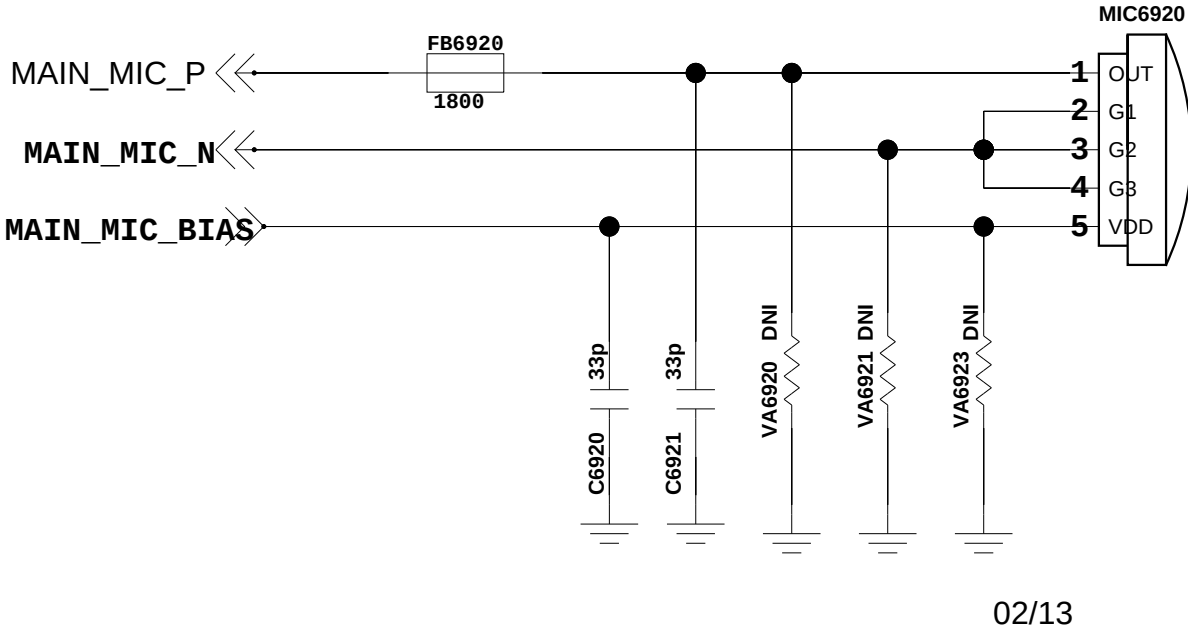


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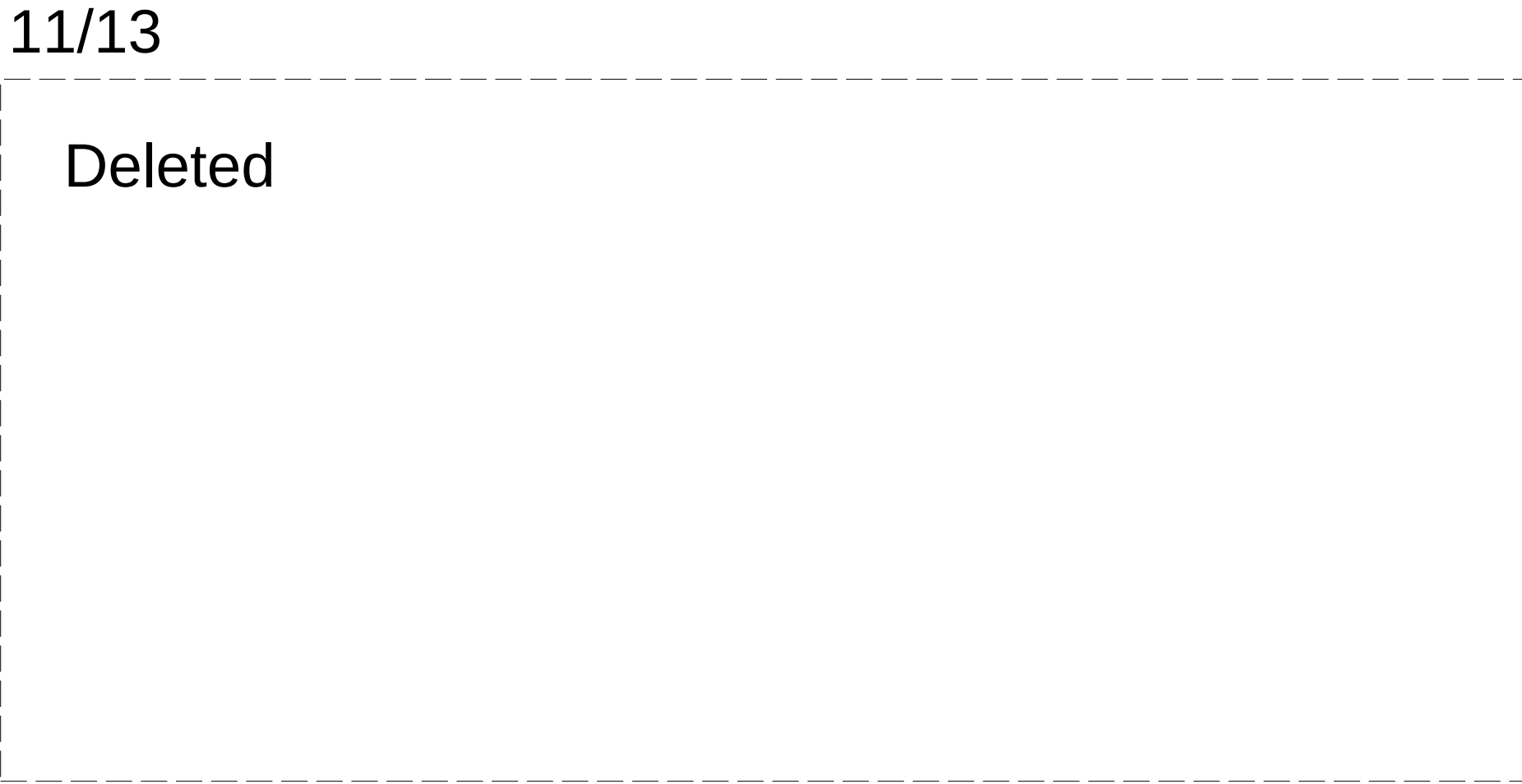


A

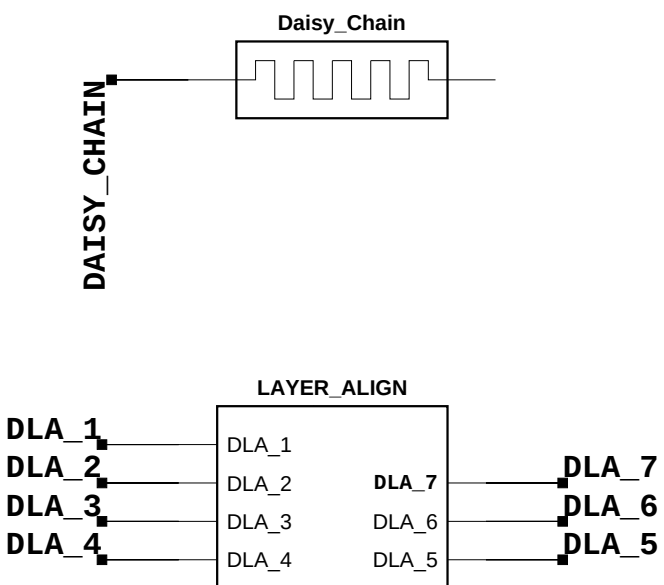
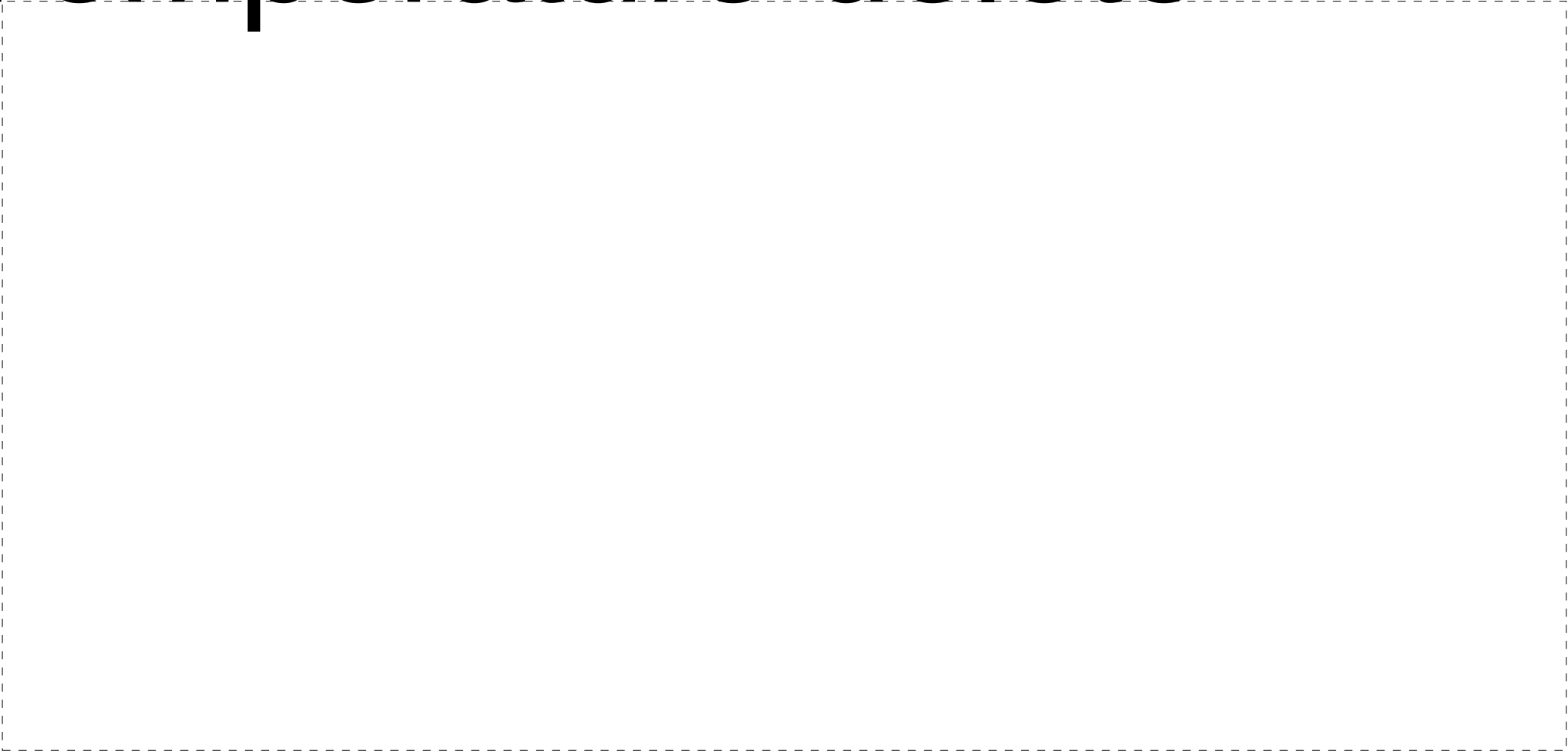
From previously Lower FPCB to Main PCB (11/13 now)



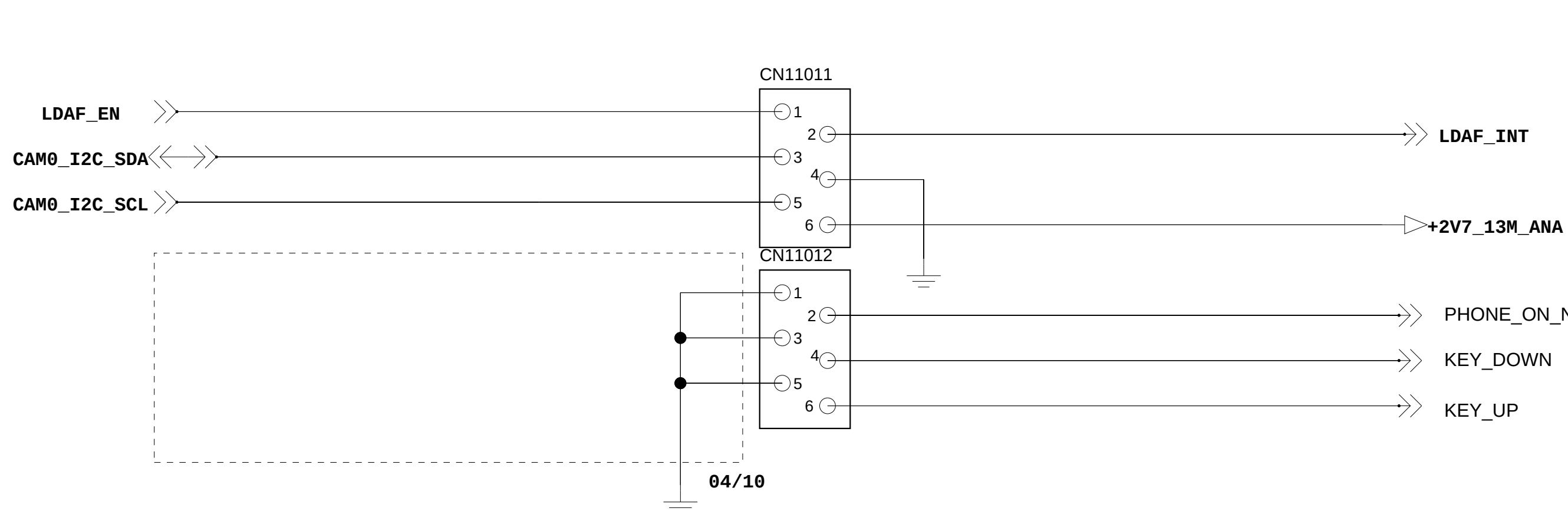
RGB LED



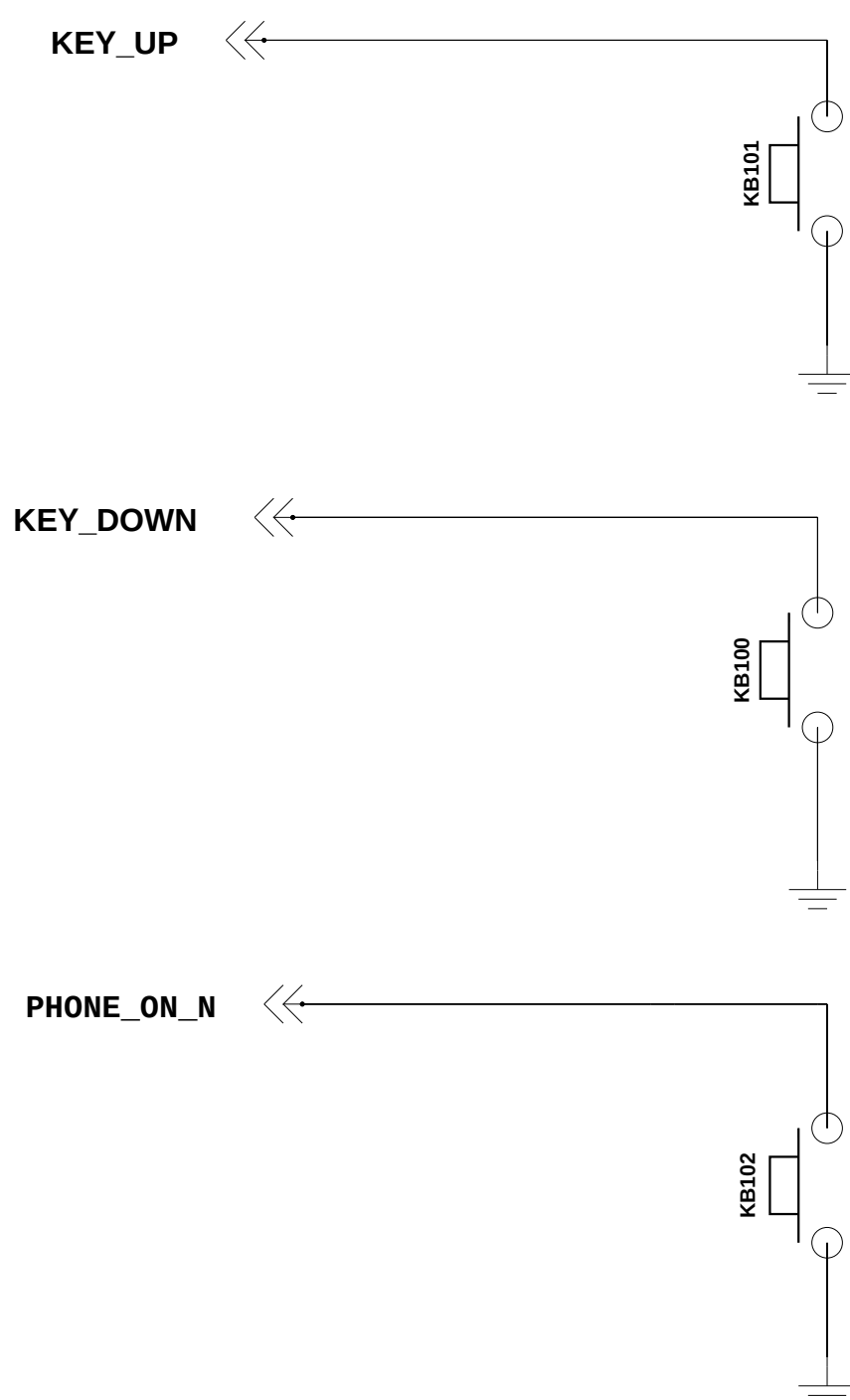
Temperature delete



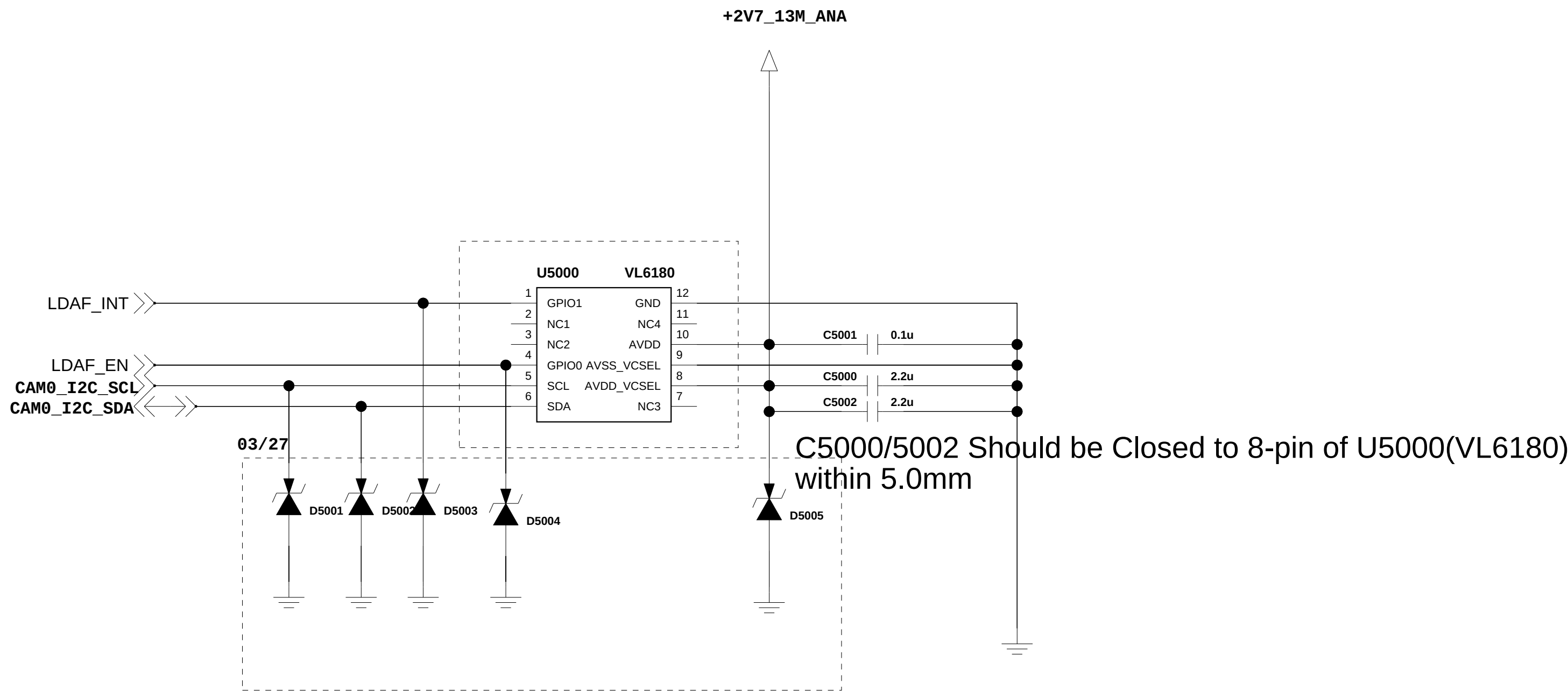
<BACK_KEY_FPCB>



<BACK_KEY>



<LDAF_VL6180>



<RGB_KEY_LED>

